TM Forum Specification

Risk Management

TMF696

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Introduction

The Risk Management API allows the retrieval of Risk Assessments during interaction with PartyRoles (individuals or organizations) typically engaged in an Order process from ProviderPartyRole. The Risk Assessment request is usually issued by the ProviderPartyRole (e.g. Service Provider). The resources used to retrieve these assessment are Task resources, as per the Open API guidelines for tasks.

Order related interactions of a PartyRole with a Service Provider entity is done via: Customer Service Representative, Call Center, Store, Dealer, Point of Sale, WEB Portal, Applications Portal, USSD Portal, etc.

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Assessed Entities

Order process related Risk Scores are assessed for the following entities:

- **PartyRole** – individual or organization playing (or intending to play) a specific role with the service provider
- **ProductOffering** – a specific offering (not necessarily related with a specific PartyRole)
- **PartyRoleProductOffering** – a specific offering addressed by the PartyRole (e.g. currently browsed/viewed by the PartyRole)
- **ShoppingCart** – the shopping cart assembled by the PartyRole
- **ProductOrder** – the overall order bound for payment by the PartyRole

As the Order Process progresses, the ProviderPartyRole may issue Risk Assessment Scores request for each of the above-mentioned entities

Following a response to a Risk Assessment request, the ProviderPartyRole may practice (its own) relevant Policies, for example:

- getting of risk assessment scores for a PartyRole may be require a mandatory request for risk scores for PartyRoleOrder (of the same PartyRole).
- getting of risk assessment scores for a ShoppingCart may result in request for an advanced full payment, rather than offer installment.

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Risk Assessment Categorizations

The following table introduces the types of risks per entities:

Scored Entity	Risk Type	Impact
PartyRole	IDConfidenceRisk	PartyRole might not be genuine (e.g. Identity Theft)
	FraudRisk	PartyRole predicted to practice fraud
	BadPaymentRisk	PartyRole predicted to avoid payments
	CreditGamingRisk	PartyRole predicted to engage in credit gaming
ProductOffering	FraudRisk	ProductOffering predicted to be affected by fraud
	BadPaymentRisk	ProductOffering predicted to be inflicted by payment issues
	CreditGamingRisk	ProductOffering predicted to be inflicted by credit gaming
PartyRoleProductOffering	IDConfidenceRisk	PartyRole might not be genuine (e.g. Identity Theft)
	FraudRisk	PartyRoleProductOffering predicted to be affected by fraud
	BadPaymentRisk	PartyRoleProductOffering predicted to be inflicted by payment issues
	CreditGamingRisk	PartyRoleProductOffering predicted to be inflicted by credit gaming
ShoppingCart	IDConfidenceRisk	PartyRole might not be genuine (e.g. Identity Theft)
	FraudRisk	ShoppingCart predicted to be affected by fraud
	BadPaymentRisk	ShoppingCart predicted to be inflicted by payment issues
	CreditGamingRisk	ShoppingCart predicted to be inflicted by credit gaming
ProductOrder	IDConfidenceRisk	PartyRole might not be genuine (e.g. Identity Theft)
	PaymentMethodRisk	PartyRole might use “questionable” payment credentials (e.g. too many/suspicious purchases with a credit card)
	FraudRisk	ProductOrder predicted to be affected by fraud
	BadPaymentRisk	ProductOrder predicted to be inflicted by payment issues

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Scored Entity	Risk Type	Impact
	CreditGamingRisk	ProductOrder predicted to be inflicted by credit gaming

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Sample Use Cases

Assessing Risks of a PartyRole

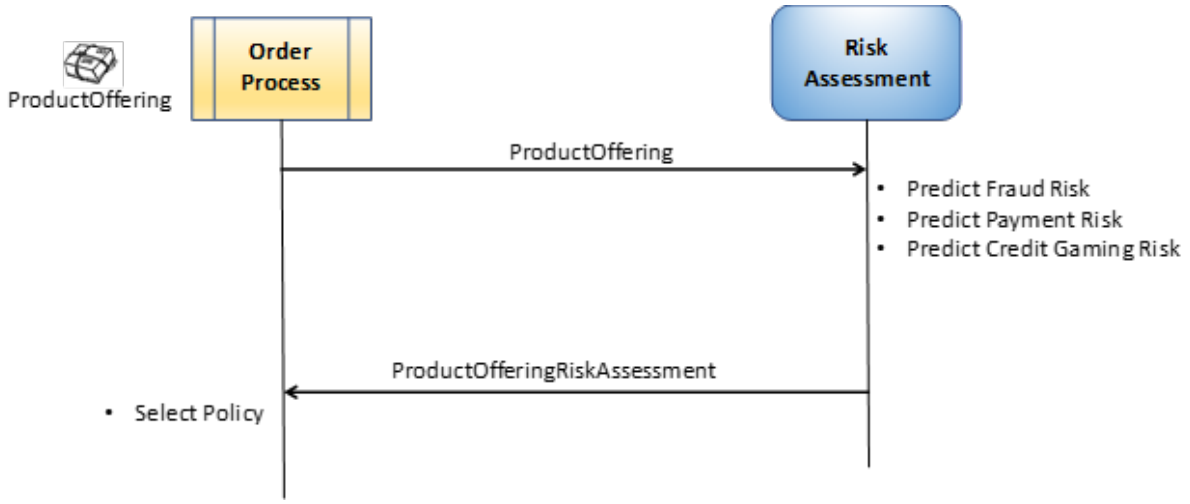
Assess Risks related with a PartyRole introduced to an Order process



According to the retrieved PartyRoleRiskAssessment, the Order process may hide specific offerings for that PartyRole. If the IDConfidenceRisk assesses a non-genuine PartyRole (e.g. account take-over or false ID credentials), the Order process may stop or request additional ID credentials.

Assessing Risks of a ProductOffering

Assess Risks related with a ProductOffering, not necessarily related with a PartyRole

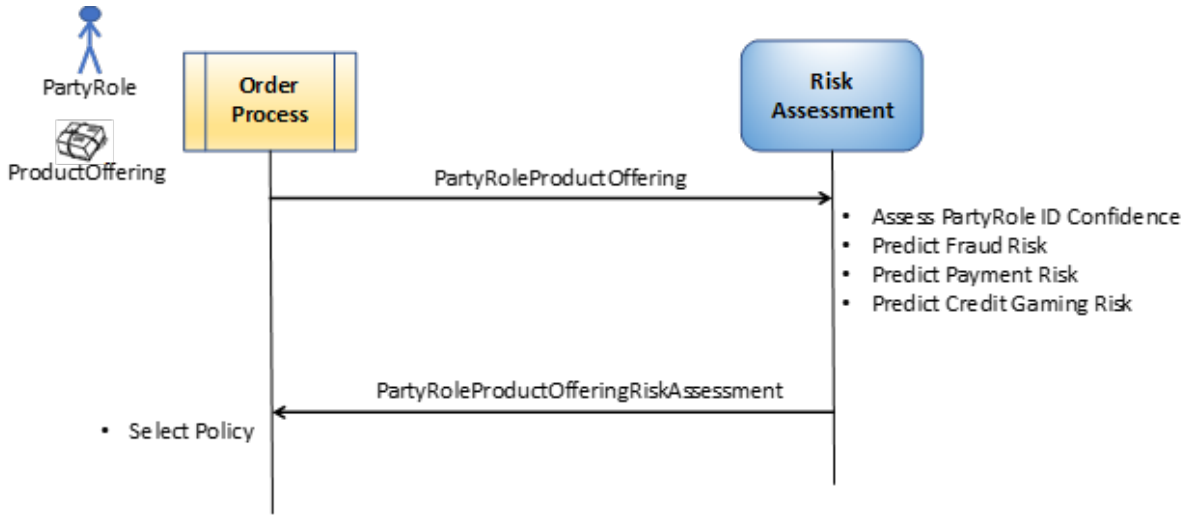


According to the retrieved ProductOfferingRiskAssessment, the Order process may choose to highlight or hide specific offerings to PartyRoles associated with specific Places.

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Assessing Risks of a PartyRoleProductOffering

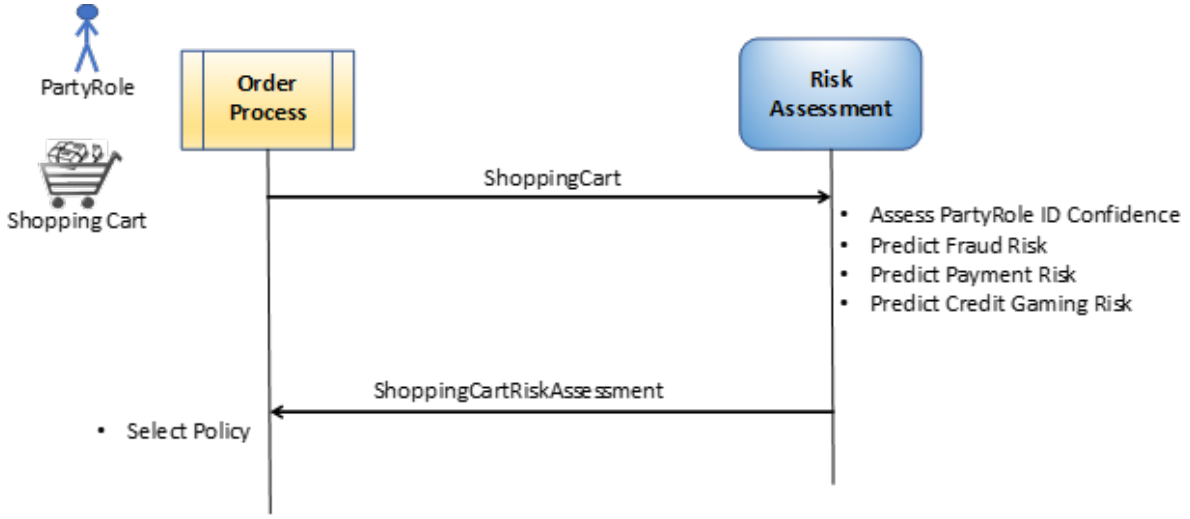
Assess Risks related with a ProductOffering currently observrd by a specific PartyRole



According to the retrieved PartyRoleProductOfferingRiskAssessment, the Order process may choose to alter the purchase criteria of that ProductOffering for that specific PartyRole.

Assessing Risks of a ShoppingCart

Assess Risks related with a ShoppingCart currently assembled by a specific PartyRole

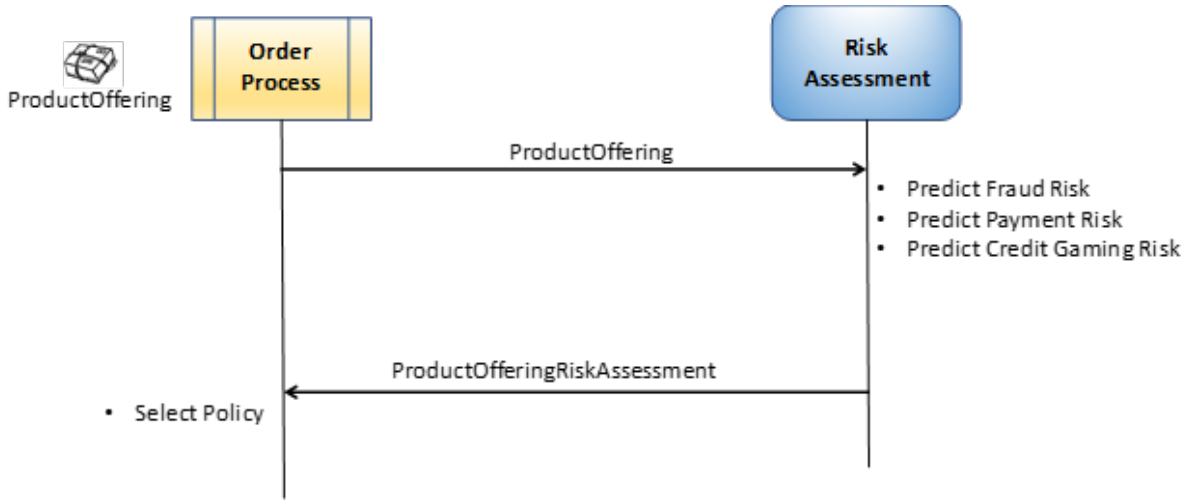


According to the retrieved ShoppingCartRiskAssessment, the Order process may choose to alter the payment terms criteria of that ShoppingCart for that specific PartyRole.

Assessing Risks of a ProductOrder

Assess Risks related with a ProductOrder currently forwarded for payment

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According to the retrieved ProductOrderRiskAssessment, the Order process may choose not to offer payment installment. If PaymentMethodRisk indicates an issue than, the Order process may request other payment credentials.

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Task Mode of Operation

The risk assessment tasks may be executed synchronously or asynchronously, according to the specific implementation.

- For an implementation that supports synchronous execution, the POST of the task is expected to return an immediate result.
  - If the implementation chooses to persist the task (e.g. for accounting purposes), then the id and href should be populated in the POST response, and the implementation must support GET and DELETE.
  - If the implementation does not persist the task, then the id and href should not be populated in the POST response, and the implementation must not support GET and DELETE.
- For an implementation that supports asynchronous execution, the POST of the task is expected to return an interim result, and to raise an event when the execution is complete. The id and href should be populated in the POST response, and the implementation must support GET and DELETE.

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema-based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v3.0 Part 2 document.

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of RiskScore instances some may be instances of an enhanced resource RiskScoreWithConfidence (with a confidence level added for the score) where other could be instances of the basic RiskScore. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, and not the class type of the referred object.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected *characteristics* of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended, so if @type was RiskScoreWithConfidence then @baseType would be RiskScore.

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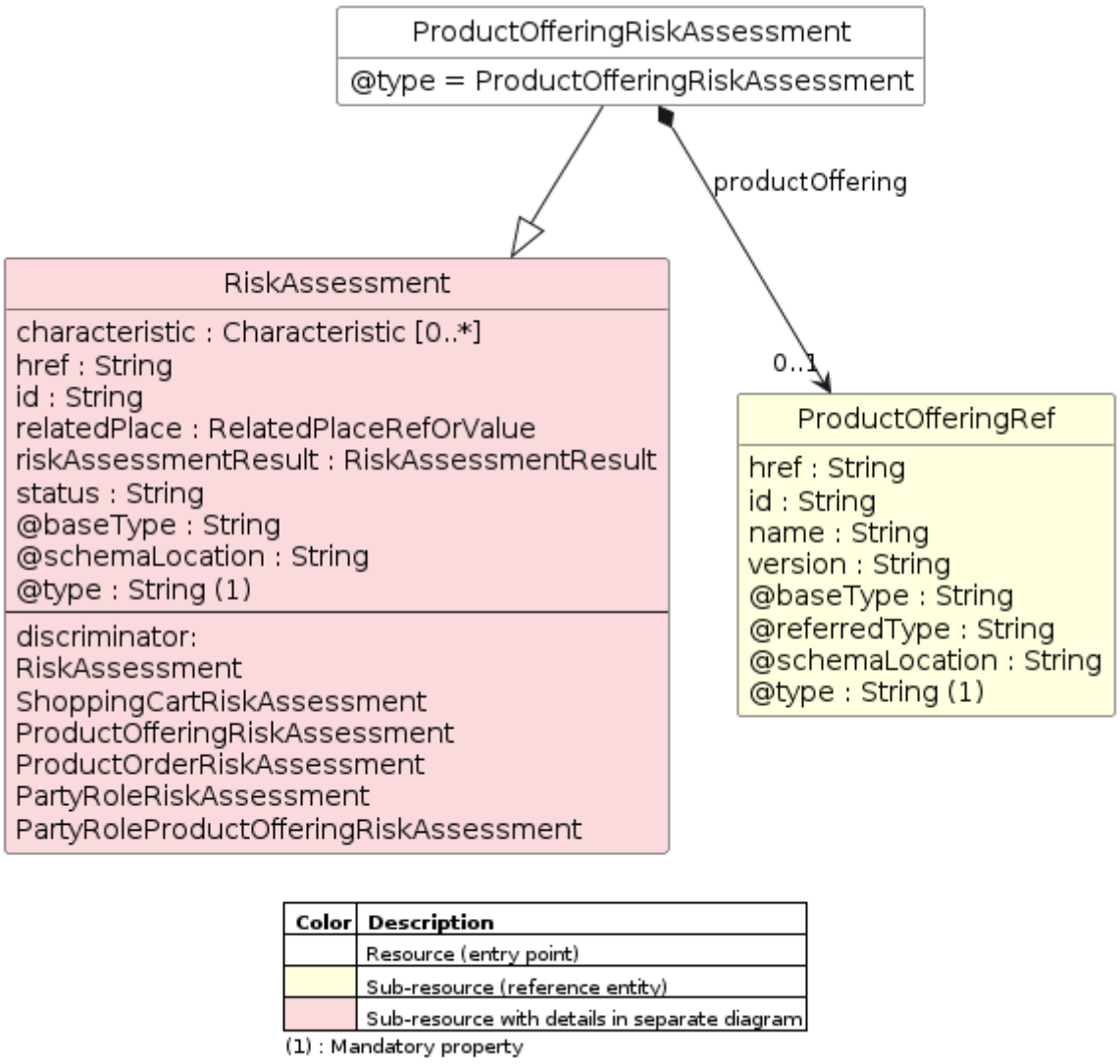
RESOURCE MODEL

Managed Entity and Task Resource Models

ProductOfferingRiskAssessment resource

This is a task resource allowing assessment of the risks associated with a product offering. For example, the risk associated with iPhone X will probably be higher than the risk for a battery.

Resource model



Color	Description
	Resource (entry point)
	Sub-resource (reference entity)
	Sub-resource with details in separate diagram

(1) : Mandatory property

Figure 1 - ProductOfferingRiskAssessment

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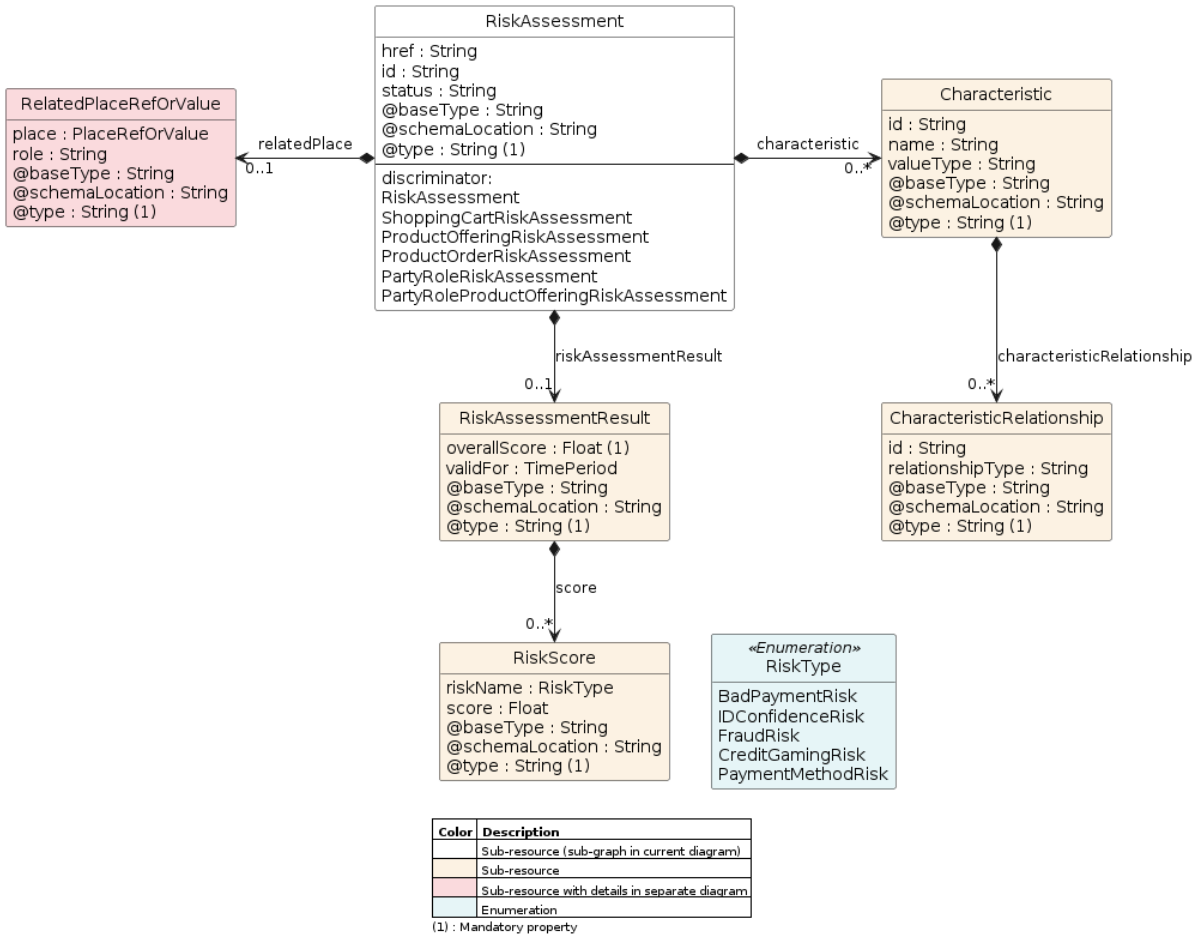


Figure 2 - RiskAssessment

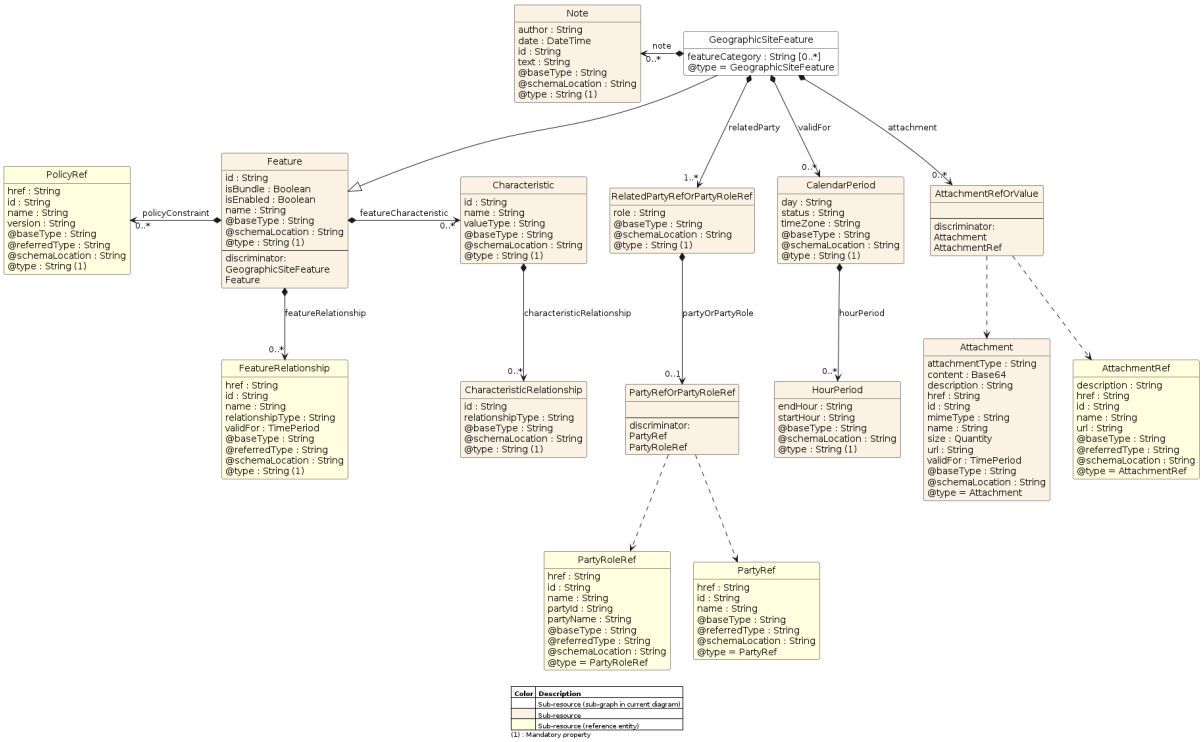


Figure 3 - GeographicSiteFeature

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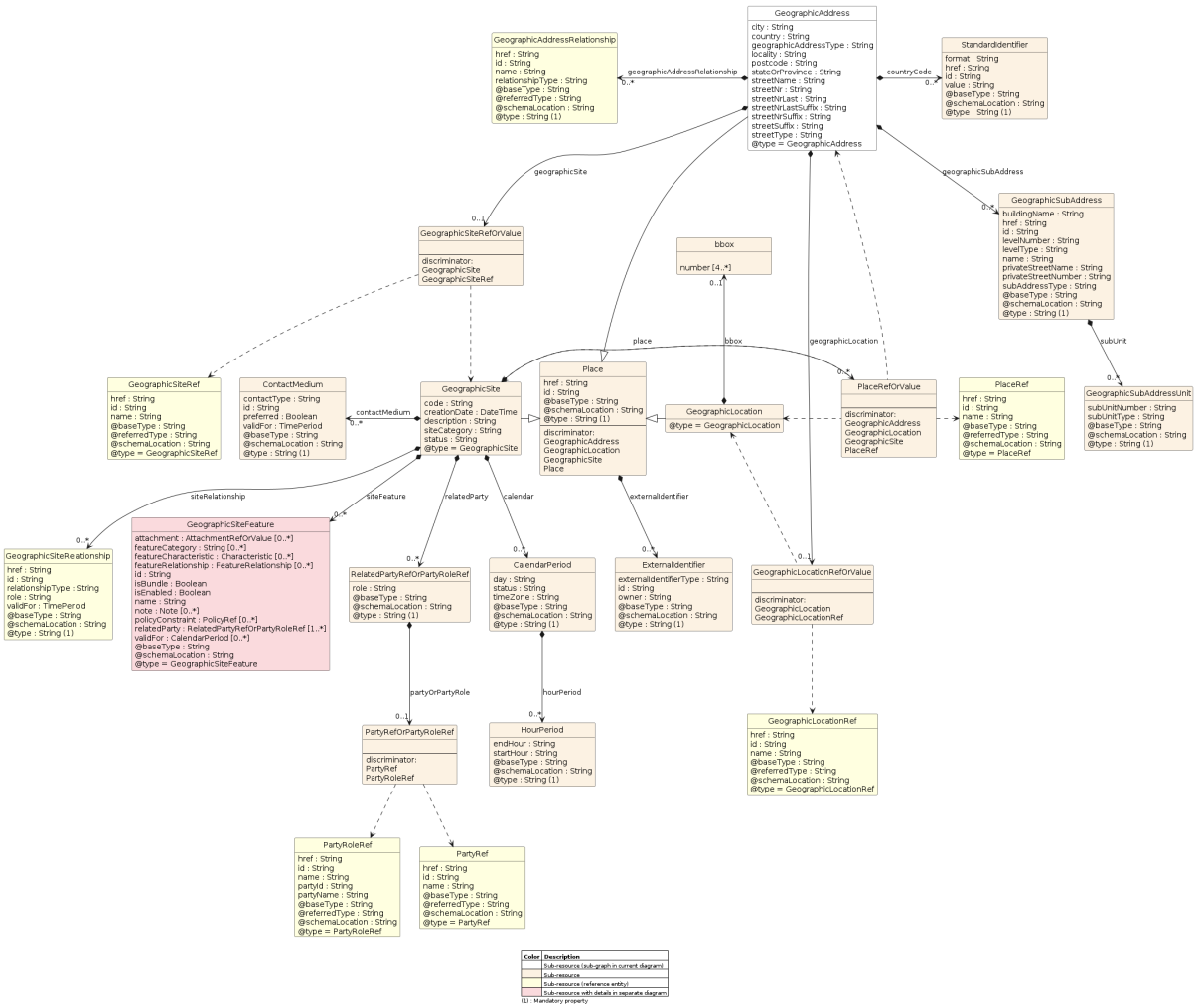


Figure 4 - GeographicAddress

Field descriptions

ProductOfferingRiskAssessment fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.

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relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the product offering.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.

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riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	RiskAssessment can be instanciated as * PartyRoleProductOfferingRiskAssessment * PartyRoleRiskAssessment * ProductOfferingRiskAssessment * ProductOrderRiskAssessment * ShoppingCartRiskAssessment
partyRole	This property is present in subclasses
productOffering	This property is present in subclasses
productOrder	This property is present in subclasses
shoppingCart	This property is present in subclasses

ShoppingCartRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
shoppingCart	A ShoppingCartRef. ShoppingCart (ShoppingCartRef) . The shopping cart which the recommendation is related with.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOrderRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOrder	A ProductOrderRef. A reference to an existing Product Order.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

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relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleProductOfferingRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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Json representation sample(s)

We provide below a JSON representation as example of the 'ProductOfferingRiskAssessment' resource object.


```
{
    "@type": "ProductOfferingRiskAssessment",
    "status": "Completed",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "Bandwidth",
            "valueType": "string",
            "value": "500MBps"
        }
    ],
    "productOffering": {
        "id": "7656",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
        "name": "Home Gateway",
        "@referredType": "ProductOffering",
        "@type": "ProductOfferingRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 19,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "@type": "RiskScore",
                "score": 20,
                "riskName": "FraudRisk"
            },
            {
                "@type": "RiskScore",
                "score": 7,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 5,
                "riskName": "CreditGamingRisk"
            }
        ]
    }
```


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```
}
```

PartyRoleRiskAssessment resource

This is a task resource allowing assessment of the risks associated with a party role.

Resource model

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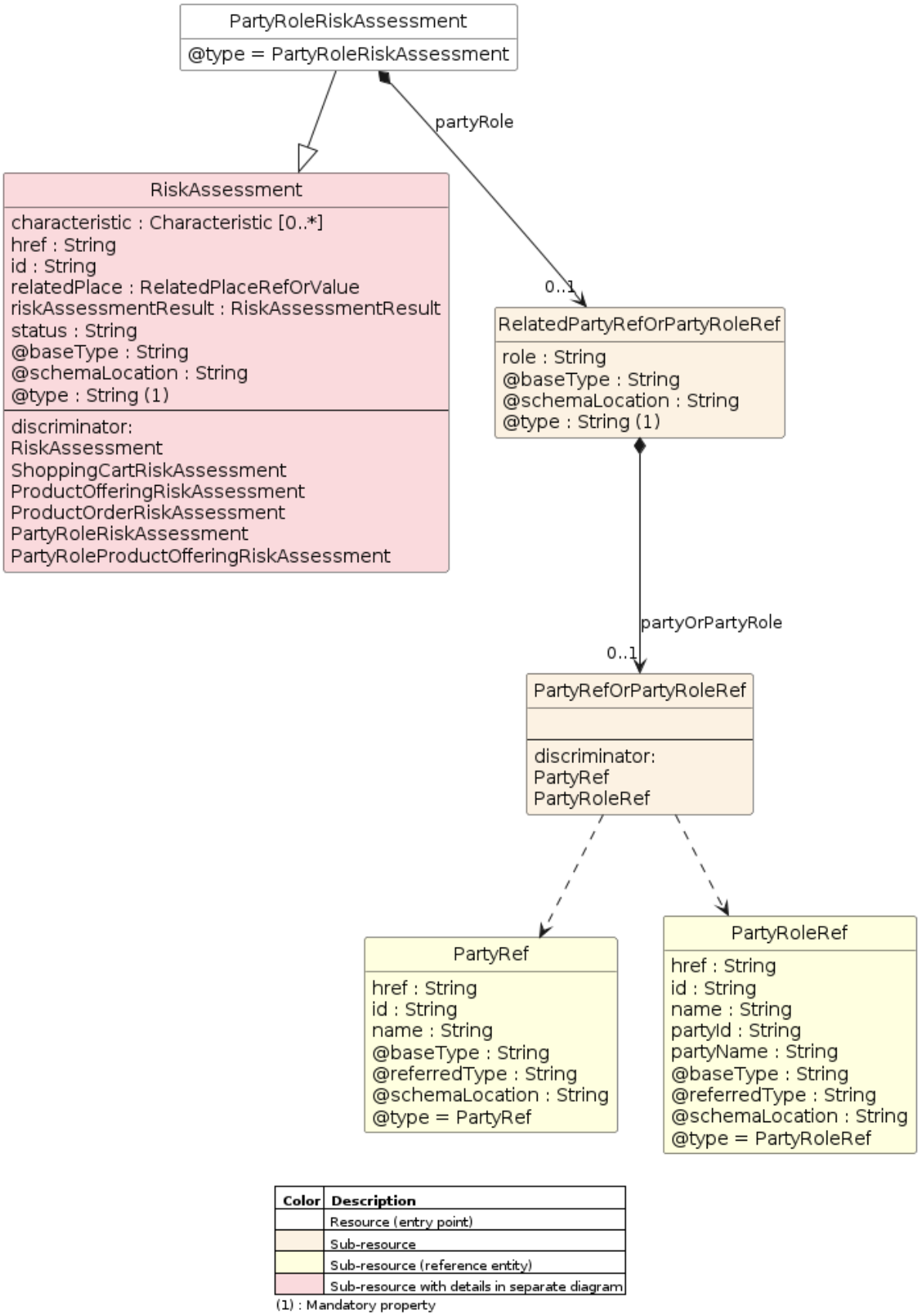


Figure 5 - PartyRoleRiskAssessment

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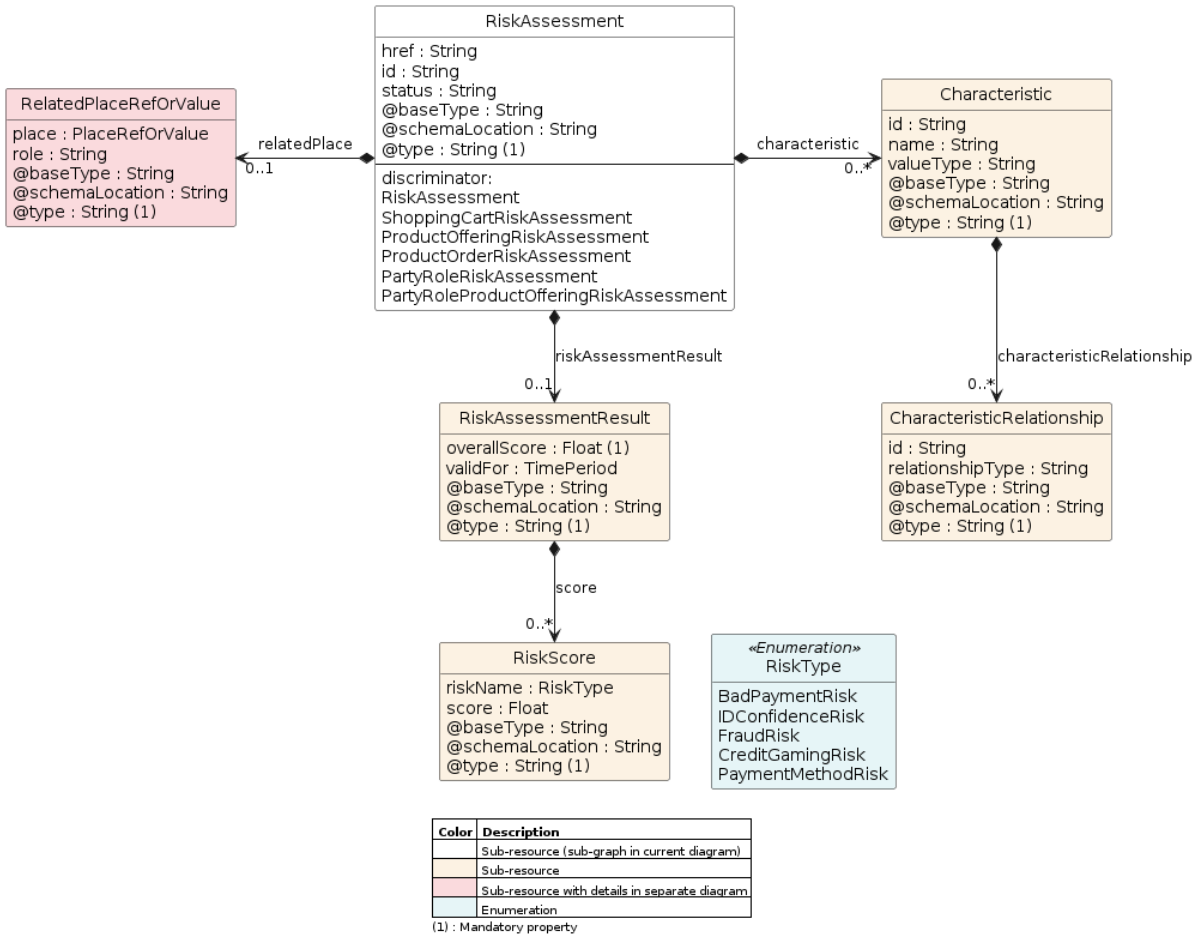


Figure 6 - RiskAssessment

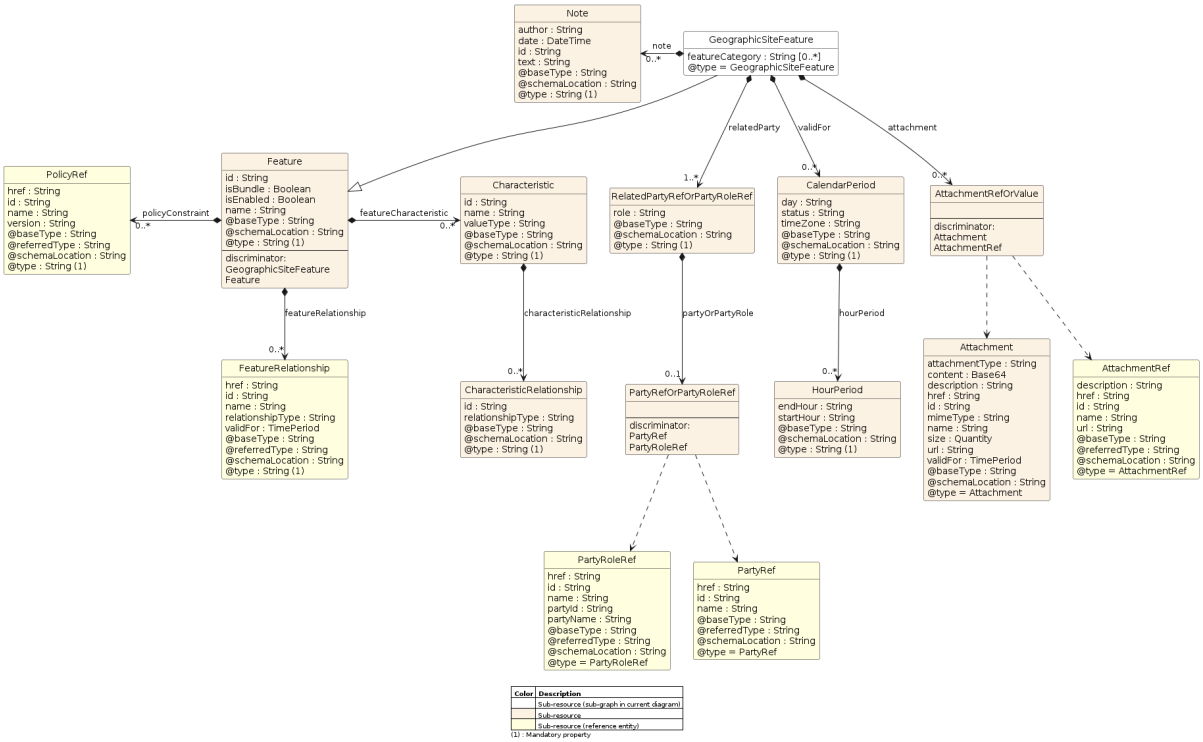


Figure 7 - GeographicSiteFeature

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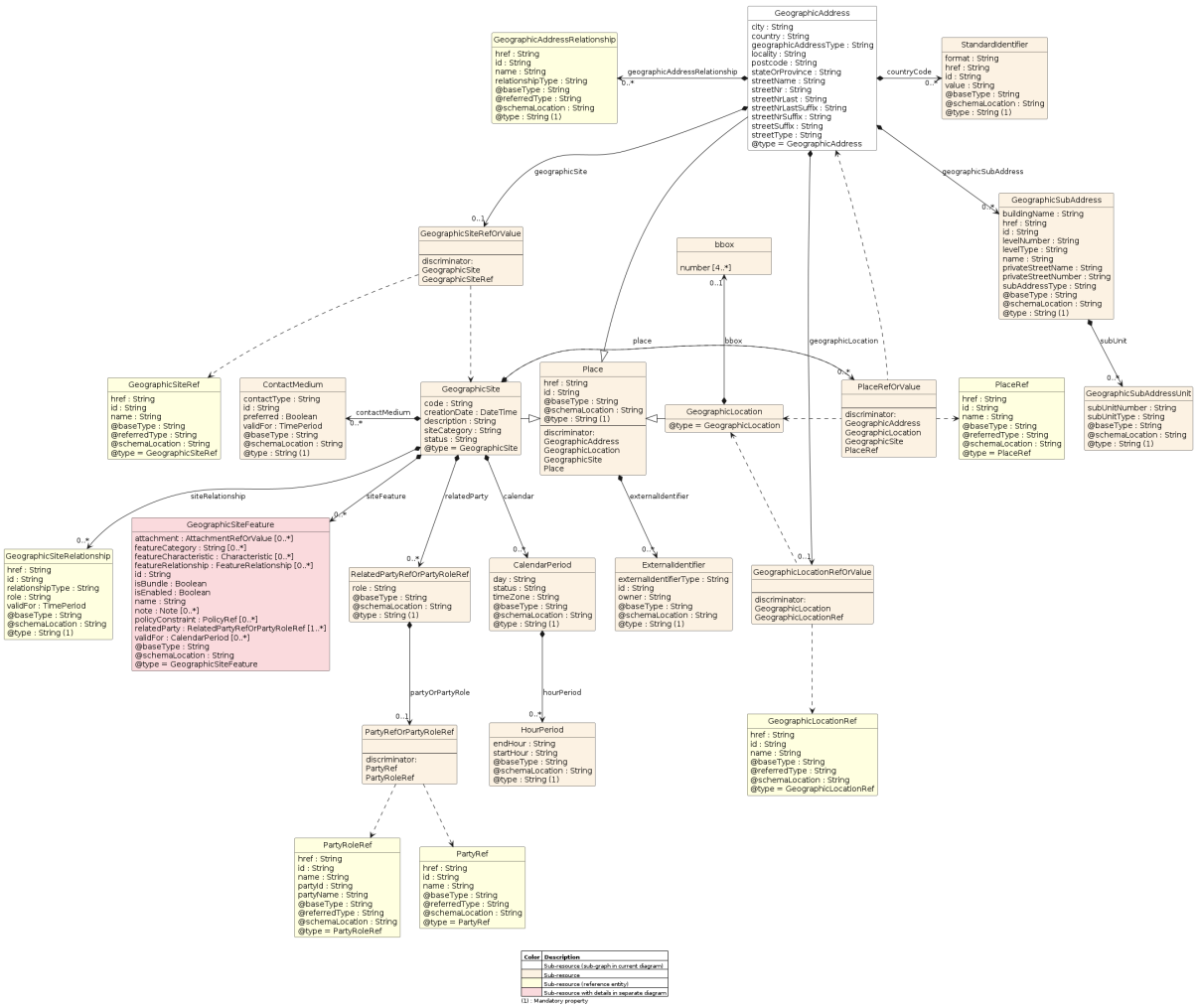


Figure 8 - GeographicAddress

Field descriptions

PartyRoleRiskAssessment fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.

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relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.

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name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	RiskAssessment can be instanciated as * PartyRoleProductOfferingRiskAssessment * PartyRoleRiskAssessment * ProductOfferingRiskAssessment * ProductOrderRiskAssessment * ShoppingCartRiskAssessment
partyRole	This property is present in subclasses

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productOffering	This property is present in subclasses
productOrder	This property is present in subclasses
shoppingCart	This property is present in subclasses

ShoppingCartRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
shoppingCart	A ShoppingCartRef. ShoppingCart (ShoppingCartRef) . The shopping cart which the recommendation is related with.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOrderRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOrder	A ProductOrderRef. A reference to an existing Product Order.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleProductOfferingRiskAssessment sub-resource fields

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characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'PartyRoleRiskAssessment' resource object.

```
{
    "@type": "PartyRoleRiskAssessment",
    "status": "Completed",
    "partyRole": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "potential buyer",
        "partyOrPartyRole": {
            "id": "9866",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
        }
    },
```

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```
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 65,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "IDConfidenceRisk"
            },
            {
                "@type": "RiskScore",
                "score": 80,
                "riskName": "FraudRisk"
            },
            {
                "@type": "RiskScore",
                "score": 20,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 14,
                "riskName": "CreditGamingRisk"
            }
        ]
    }
}
```


PartyRoleProductOfferingRiskAssessment resource

This is a task resource allowing assessment of the risks associated with a party or party role that considers acquiring a specific product offering.

Resource model

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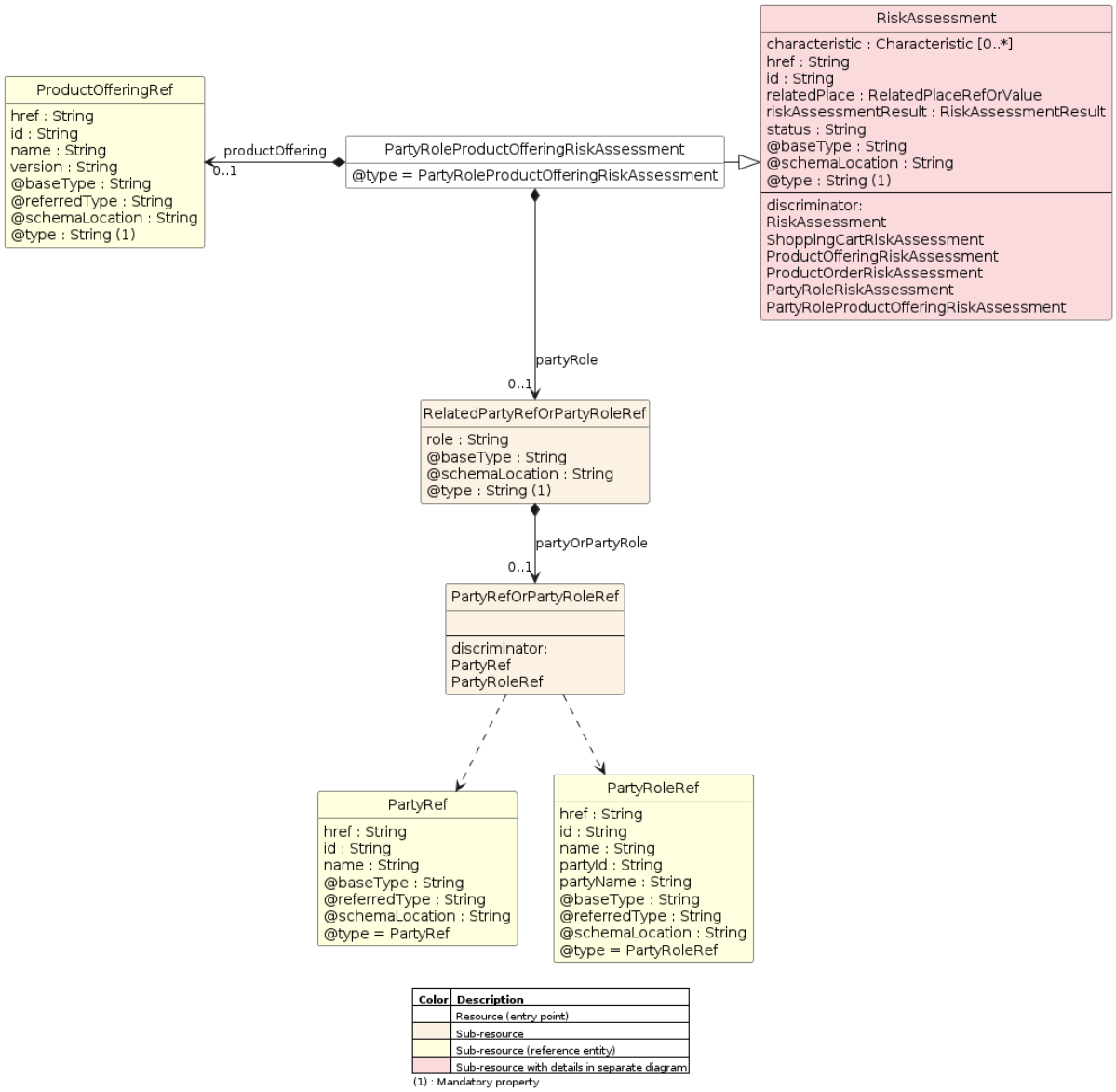


Figure 9 - PartyRoleProductOfferingRiskAssessment

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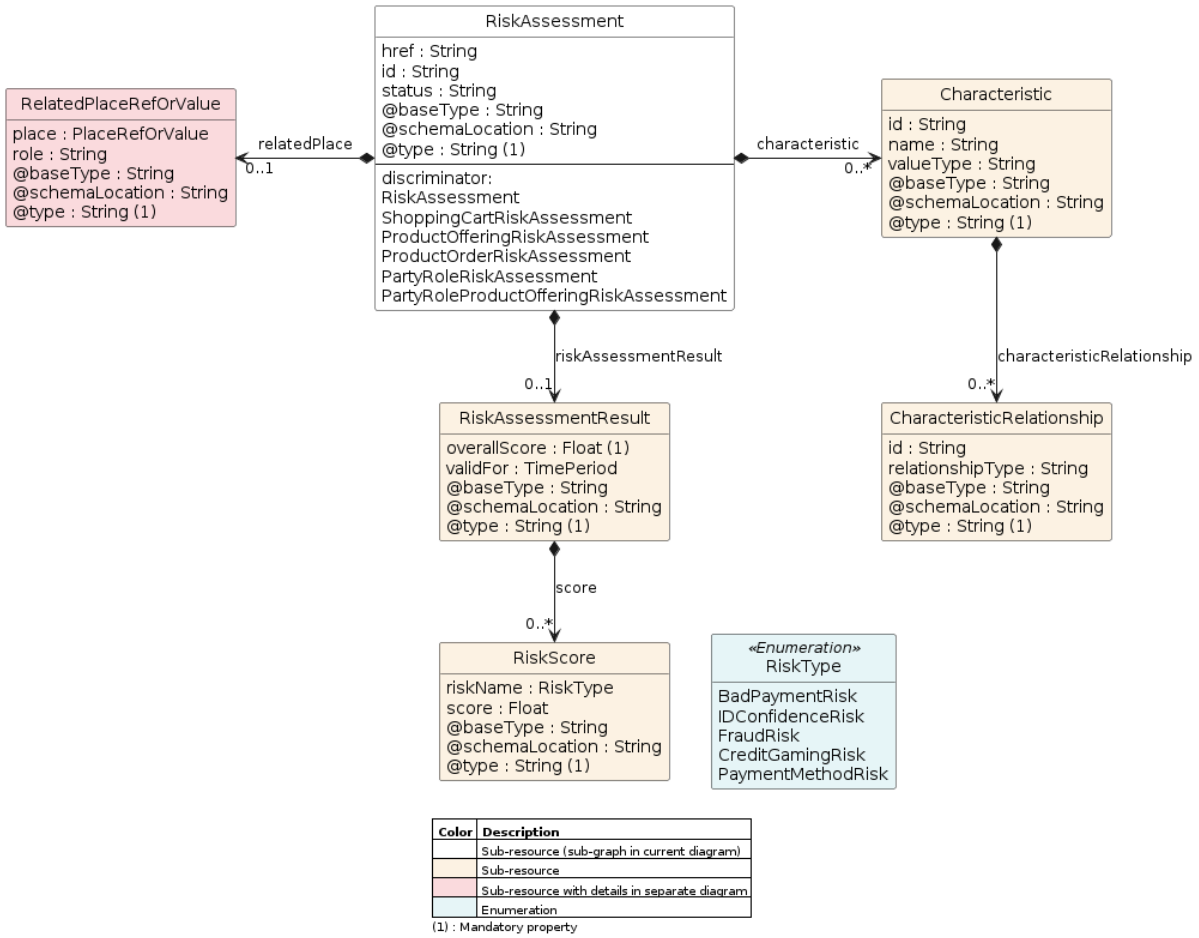


Figure 10 - RiskAssessment

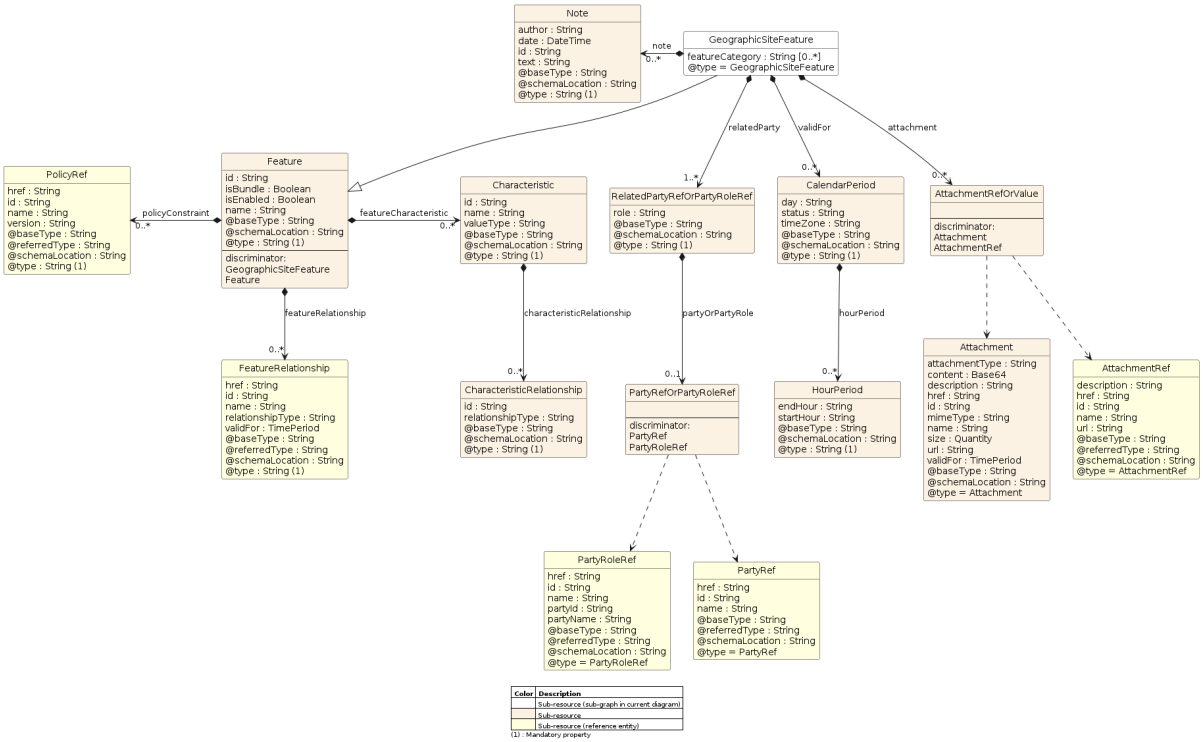


Figure 11 - GeographicSiteFeature

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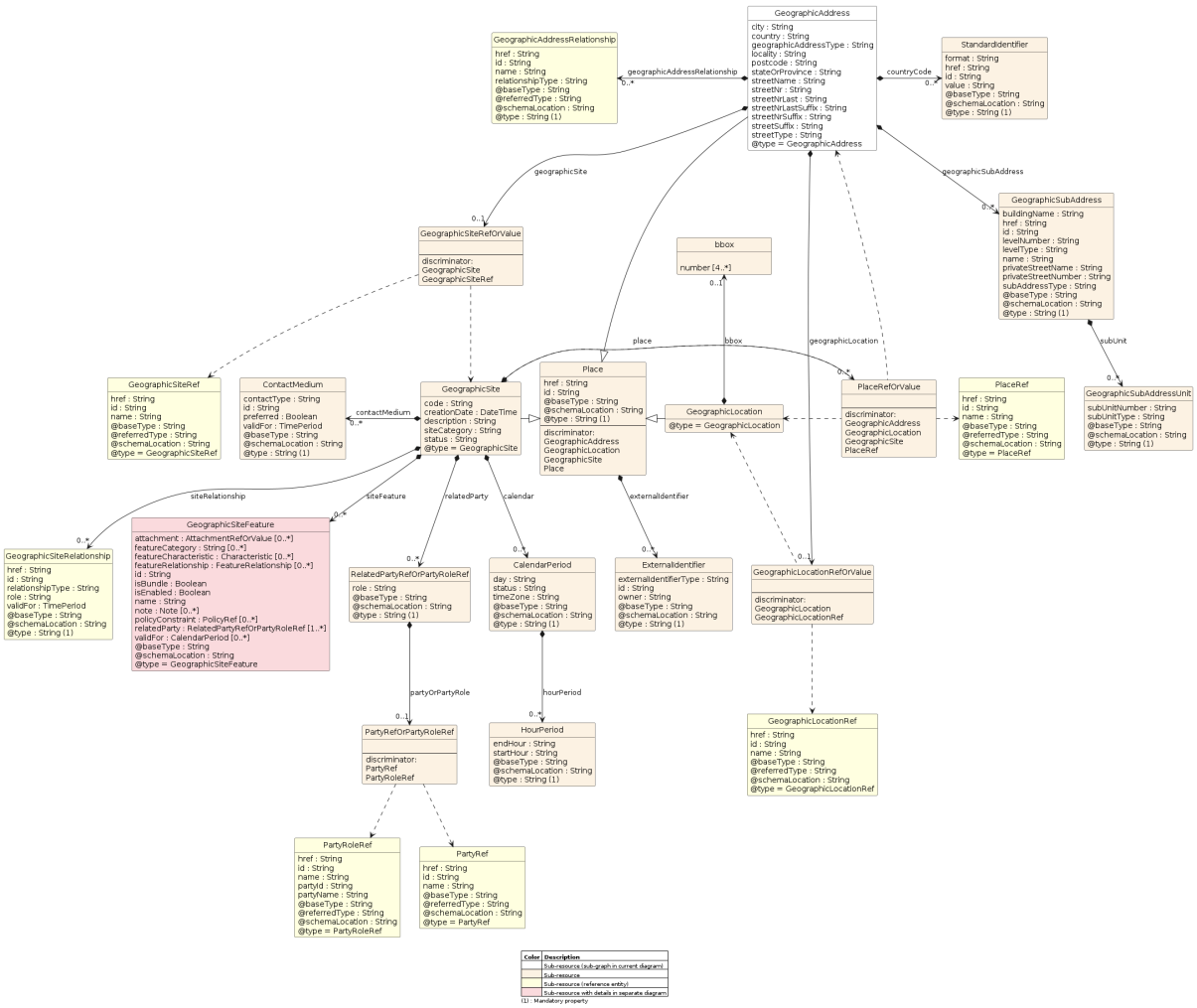


Figure 12 - GeographicAddress

Field descriptions

PartyRoleProductOfferingRiskAssessment fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.

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relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Version of the product offering.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.

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id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.

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riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	RiskAssessment can be instanciated as * PartyRoleProductOfferingRiskAssessment * PartyRoleRiskAssessment * ProductOfferingRiskAssessment * ProductOrderRiskAssessment * ShoppingCartRiskAssessment
partyRole	This property is present in subclasses
productOffering	This property is present in subclasses
productOrder	This property is present in subclasses
shoppingCart	This property is present in subclasses

ShoppingCartRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
shoppingCart	A ShoppingCartRef. ShoppingCart (ShoppingCartRef) . The shopping cart which the recommendation is related with.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOrderRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOrder	A ProductOrderRef. A reference to an existing Product Order.

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relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the

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'PartyRoleProductOfferingRiskAssessment' resource object.


```
{
    "@type": "PartyRoleProductOfferingRiskAssessment",
    "status": "Completed",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "Bandwidth",
            "valueType": "string",
            "value": "500MBps"
        }
    ],
    "partyRole": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "potential buyer",
        "partyOrPartyRole": {
            "id": "9866",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
        }
    },
    "productOffering": {
        "id": "7656",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
        "name": "Home Gateway",
        "@referredType": "ProductOffering",
        "@type": "ProductOfferingRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 21,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "IDConfidenceRisk"
            },
            {
                "@type": "RiskScore",
                "score": 32,
                "riskName": "FraudRisk"
```


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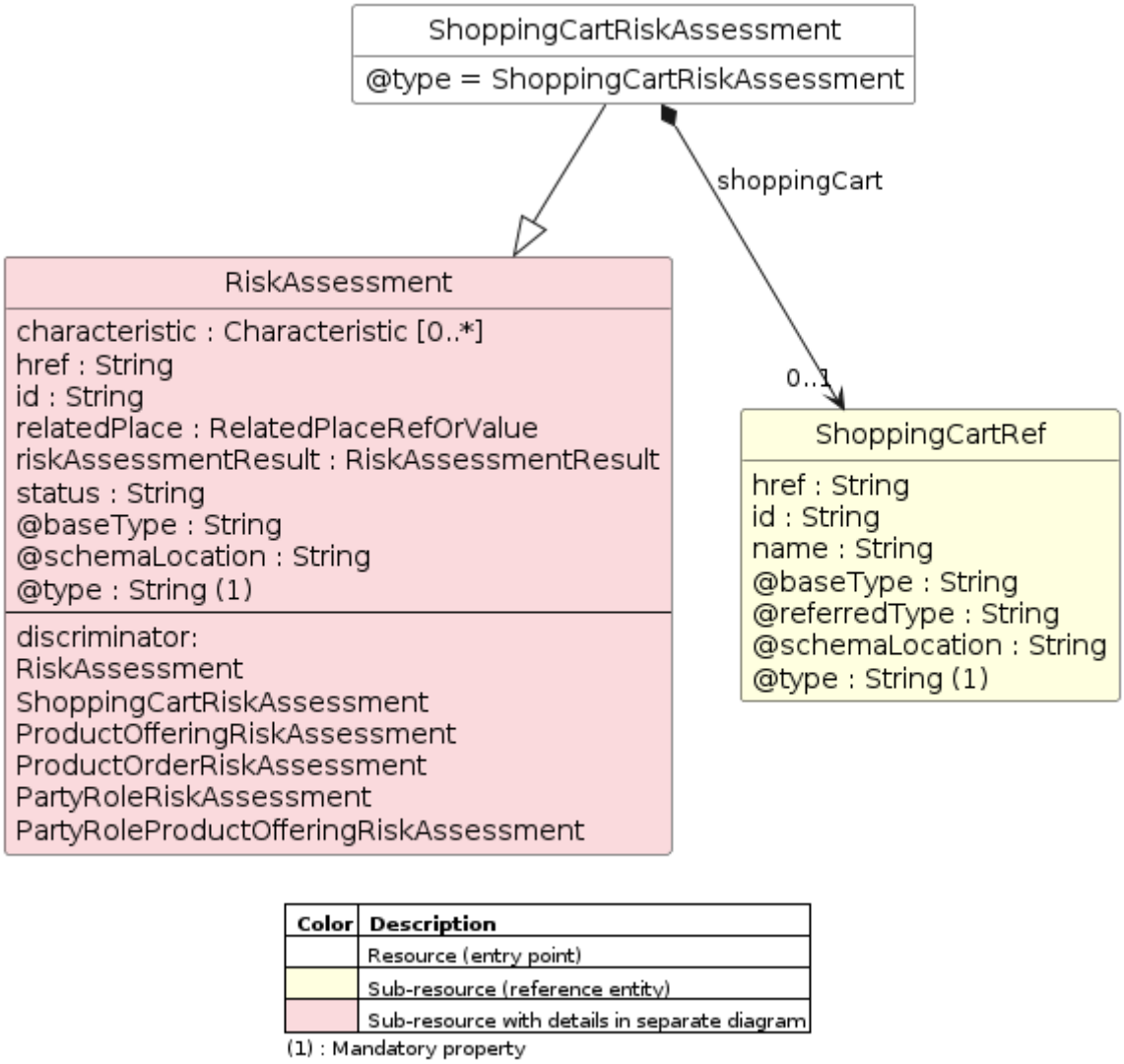
```
            },
            {
                "@type": "RiskScore",
                "score": 8,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 16,
                "riskName": "CreditGamingRisk"
            }
        ]
    }
}
```


ShoppingCartRiskAssessment resource

This is a task resource allowing assessment of the risks associated with a shopping cart - presumably created by a party role and containing one or more product offerings.

Resource model

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Color	Description
	Resource (entry point)
	Sub-resource (reference entity)
	Sub-resource with details in separate diagram

(1) : Mandatory property

Figure 13 - ShoppingCartRiskAssessment

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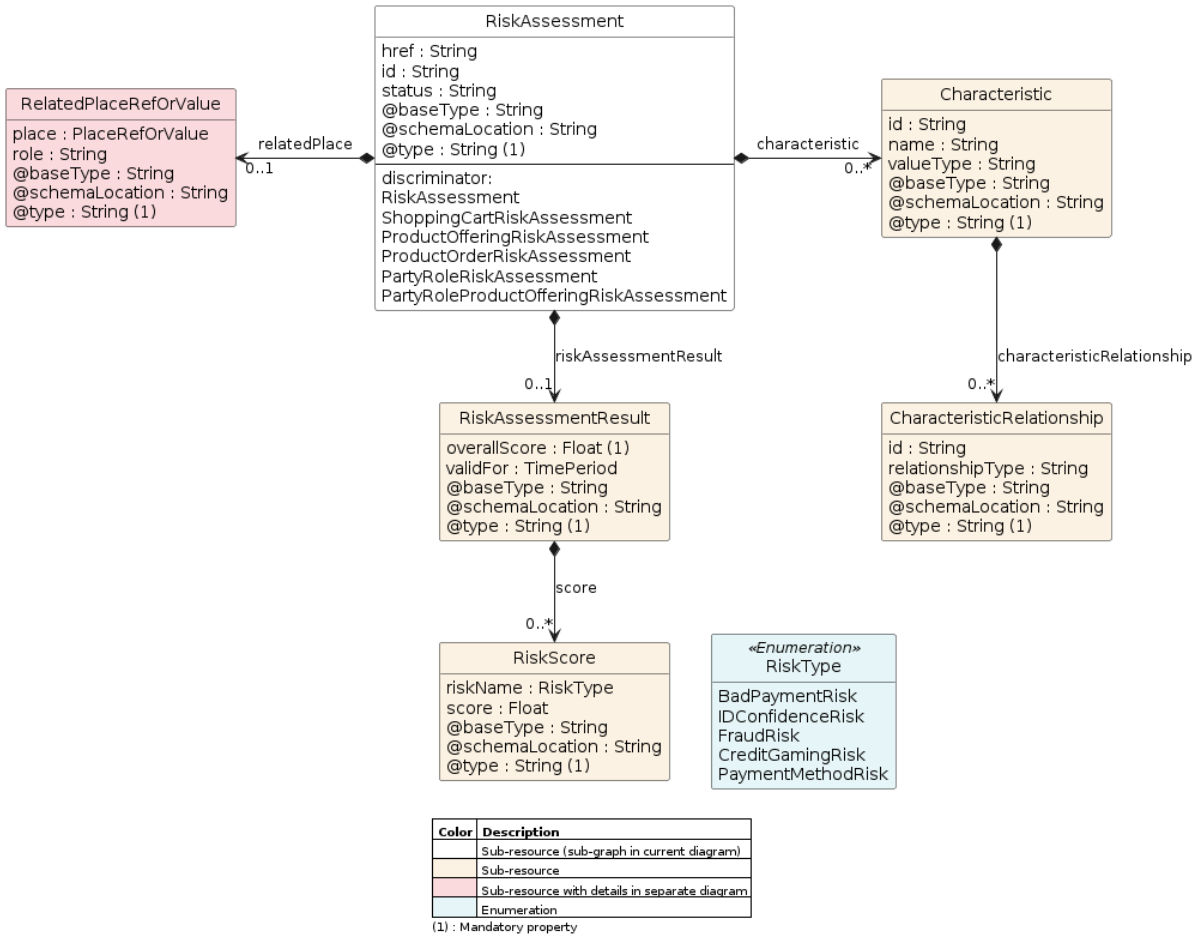


Figure 14 - RiskAssessment

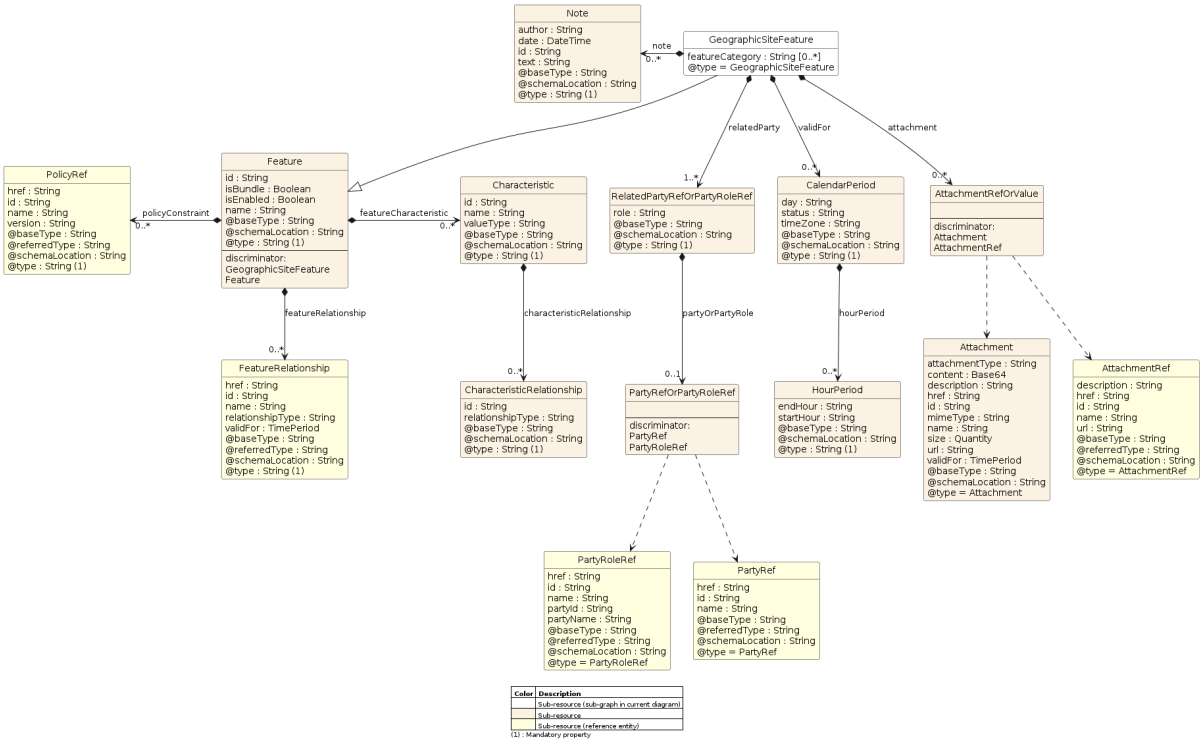


Figure 15 - GeographicSiteFeature

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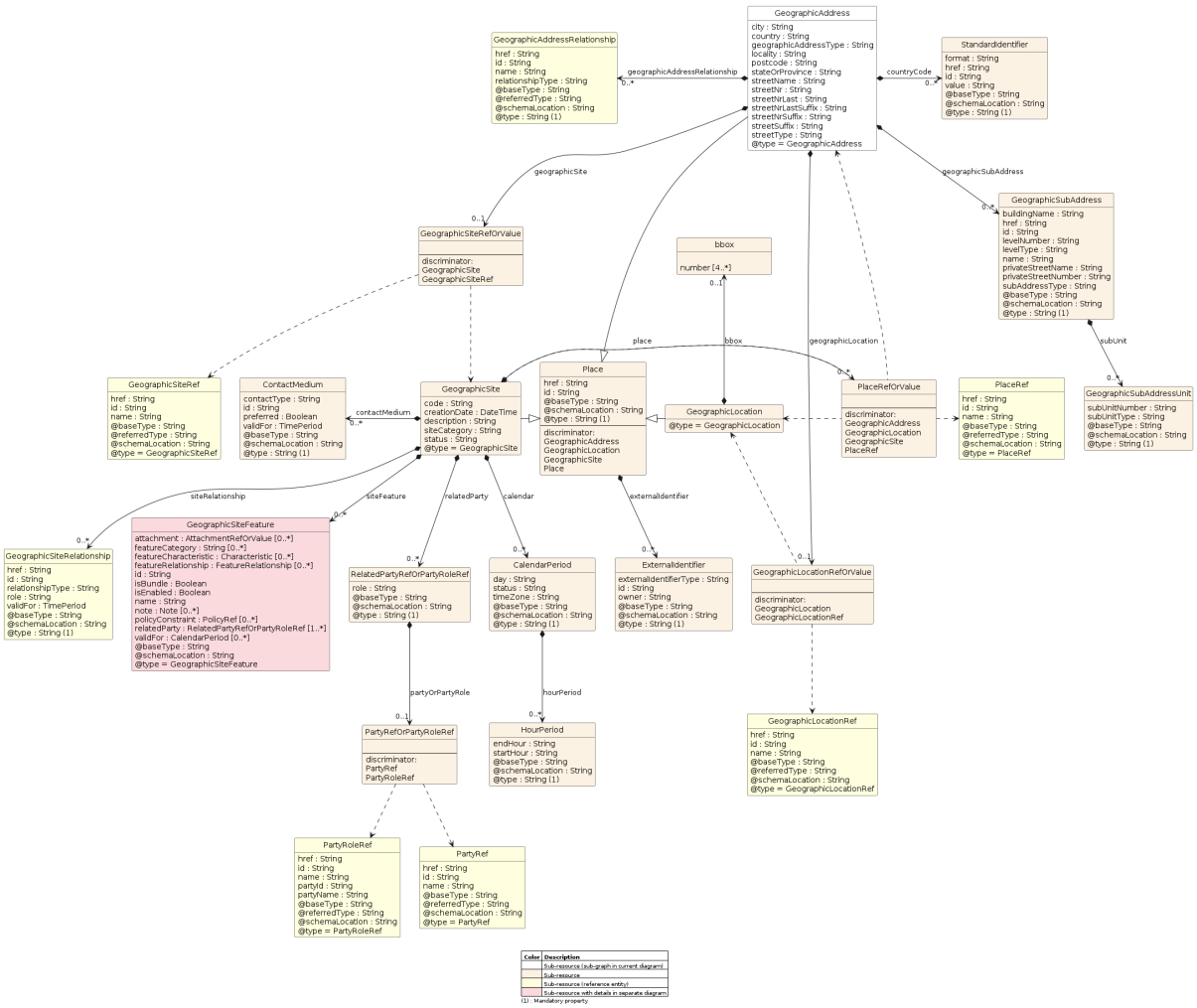


Figure 16 - GeographicAddress

Field descriptions

ShoppingCartRiskAssessment fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.

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shoppingCart	A ShoppingCartRef. ShoppingCart (ShoppingCartRef) . The shopping cart which the recommendation is related with.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ShoppingCartRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	RiskAssessment can be instanciated as * PartyRoleProductOfferingRiskAssessment * PartyRoleRiskAssessment * ProductOfferingRiskAssessment * ProductOrderRiskAssessment * ShoppingCartRiskAssessment
partyRole	This property is present in subclasses
productOffering	This property is present in subclasses
productOrder	This property is present in subclasses
shoppingCart	This property is present in subclasses

ProductOfferingRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOrderRiskAssessment sub-resource fields

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characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOrder	A ProductOrderRef. A reference to an existing Product Order.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.

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status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleProductOfferingRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ShoppingCartRiskAssessment' resource object.

```
{
```

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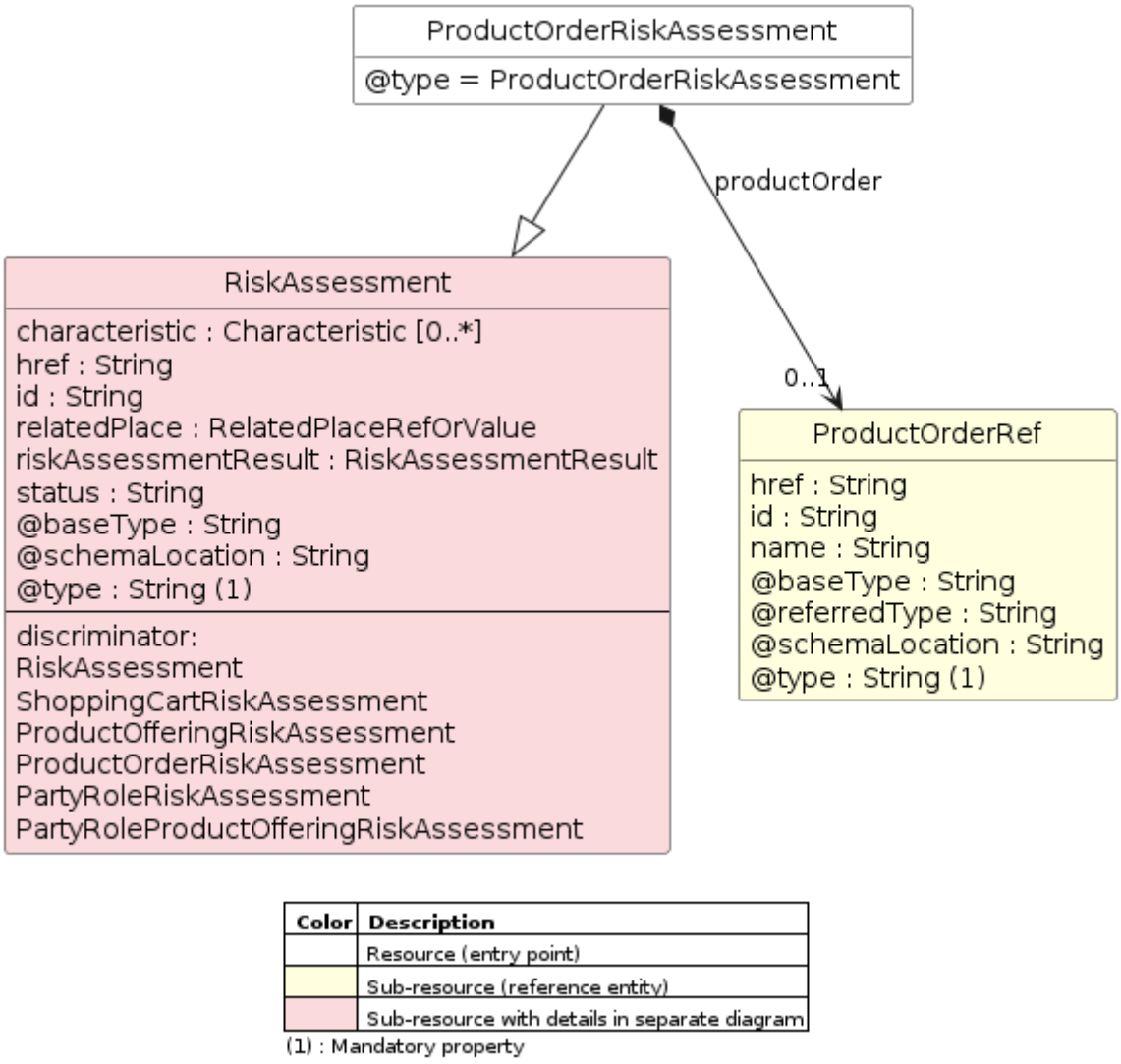
```
    "@type": "ShoppingCartRiskAssessment",
    "status": "Completed",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "ContactPersonName",
            "valueType": "string",
            "value": "David Alexander"
        }
    ],
    "shoppingCart": {
        "id": "1203",
        "href": "https://mycsp.com:8080/tmf-api/shoppingCart/v5/shoppingCart/1203",
        "@referredType": "ShoppingCart",
        "@type": "ShoppingCartRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 12,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "score": 0,
                "riskName": "IDConfidenceRisk"
            },
            {
                "score": 10,
                "riskName": "FraudRisk"
            },
            {
                "score": 13,
                "riskName": "BadPaymentRisk"
            },
            {
                "score": 9,
                "riskName": "CreditGamingRisk"
            }
        ]
    }
}
```


ProductOrderRiskAssessment resource

This is a task resource allowing assessment of the risks associated with a product order - presumably submitted by a party role and containing one or more product offerings.

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Resource model



Color	Description
	Resource (entry point)
	Sub-resource (reference entity)
	Sub-resource with details in separate diagram

(1) : Mandatory property

Figure 17 - ProductOrderRiskAssessment

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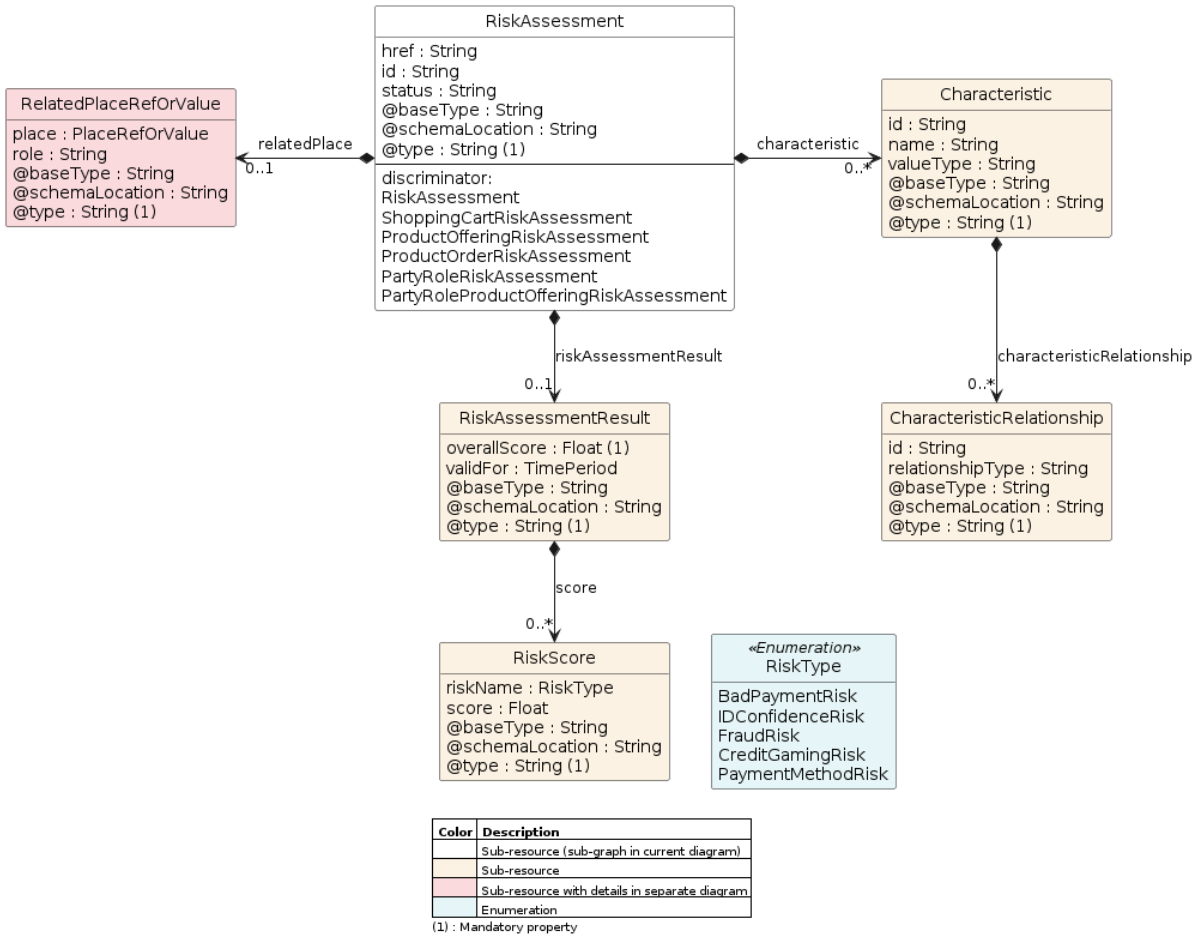


Figure 18 - RiskAssessment

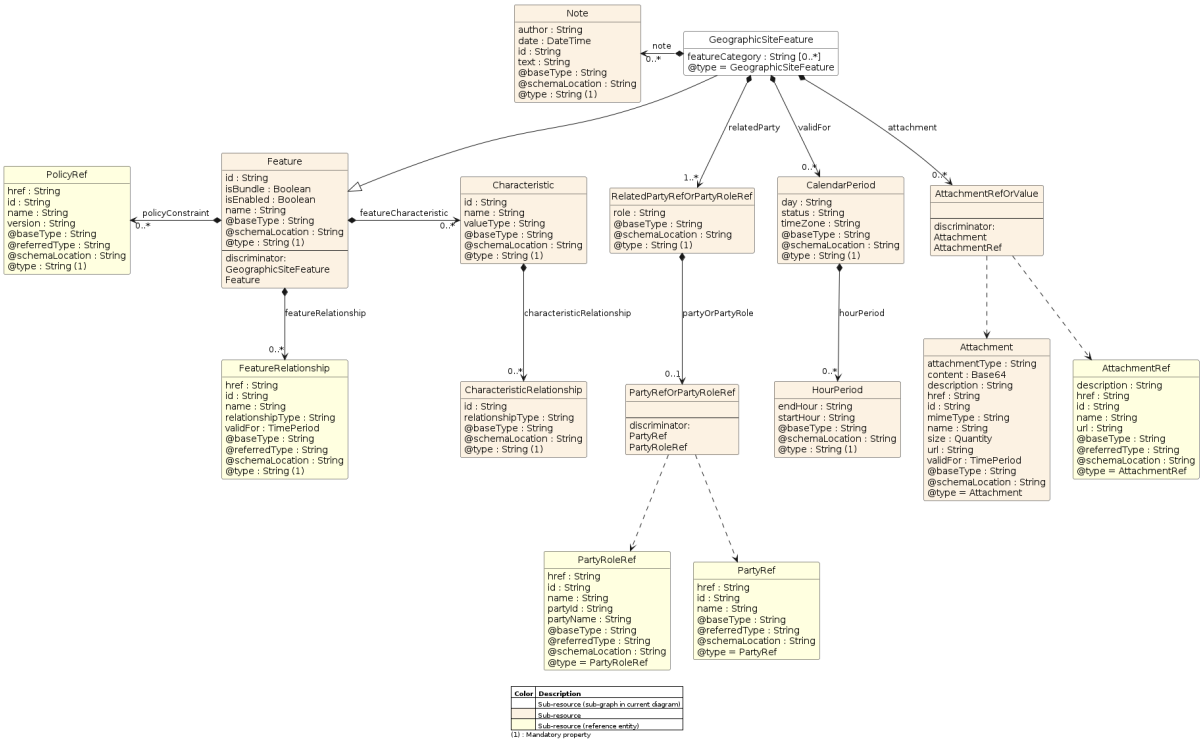


Figure 19 - GeographicSiteFeature

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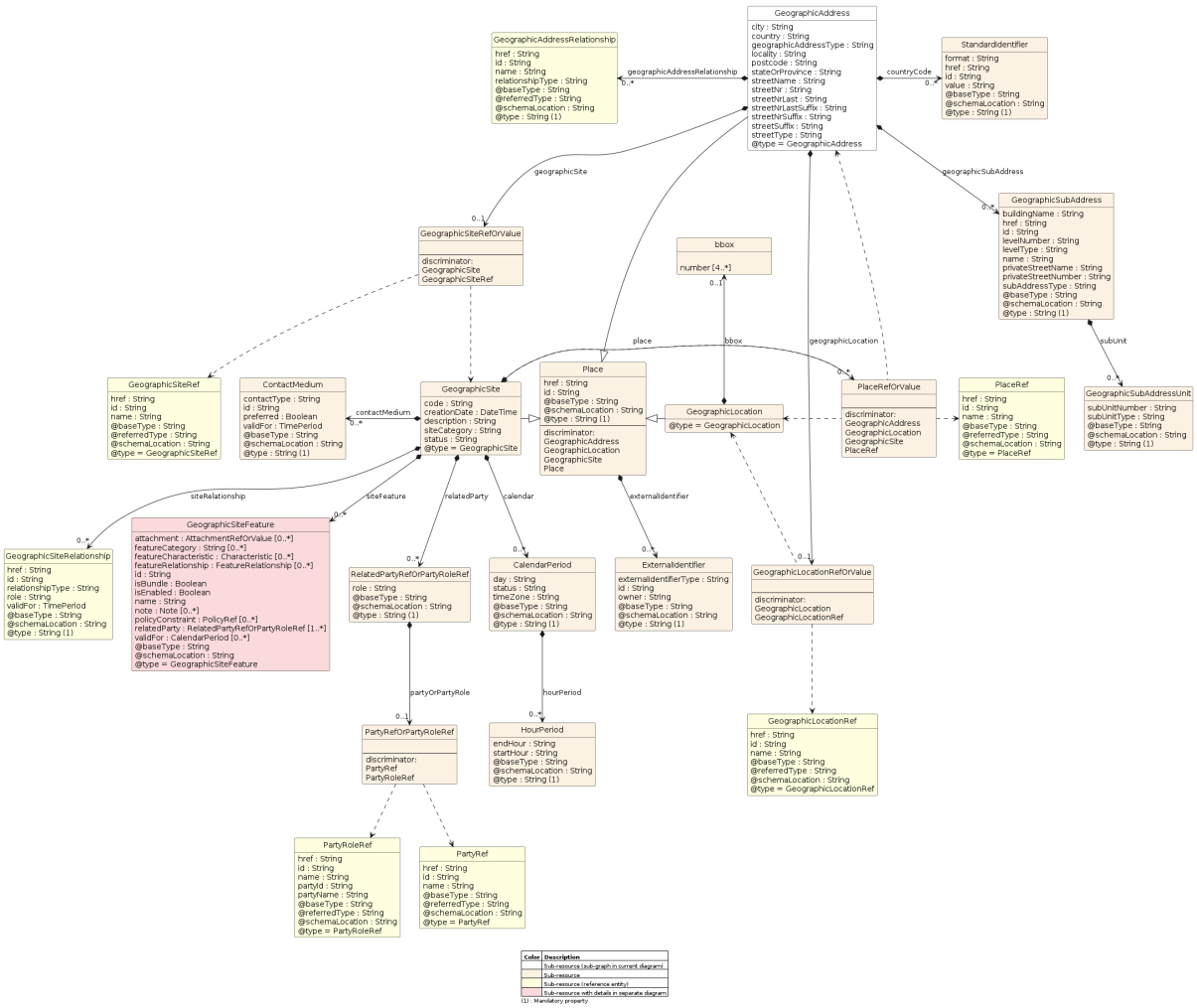


Figure 20 - GeographicAddress

Field descriptions

ProductOrderRiskAssessment fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOrder	A ProductOrderRef. A reference to an existing Product Order.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.

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riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOrderRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	RiskAssessment can be instanciated as * PartyRoleProductOfferingRiskAssessment * PartyRoleRiskAssessment * ProductOfferingRiskAssessment * ProductOrderRiskAssessment * ShoppingCartRiskAssessment
partyRole	This property is present in subclasses
productOffering	This property is present in subclasses
productOrder	This property is present in subclasses
shoppingCart	This property is present in subclasses

ShoppingCartRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
shoppingCart	A ShoppingCartRef. ShoppingCart (ShoppingCartRef) . The shopping cart which the recommendation is related with.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ProductOfferingRiskAssessment sub-resource fields

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characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.

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riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleProductOfferingRiskAssessment sub-resource fields

characteristic	A Characteristic. Describes a given characteristic of an object or entity through a name/value pair. This is an abstract base class, the actual value is in one of the strongly-typed subclasses : StringCharacteristic, ObjectCharacteristic, FloatCharacteristic, BooleanCharacteristic, NumberCharacteristic, IntegerCharacteristic, StringArrayCharacteristic, ObjectArrayCharacteristic, BooleanArrayCharacteristic, NumberArrayCharacteristic, IntegerArrayCharacteristic…
href	A String. Hyperlink reference.
id	A String. Unique identifier.
partyRole	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
productOffering	A ProductOfferingRef. ProductOffering reference. A product offering represents entities that are orderable from the provider of the catalog, this resource includes pricing information.
relatedPlace	A RelatedPlaceRefOrValue. Entity reference. The polymorphic attributes @type, @schemaLocation & @referredType are related to the RelatedPlace entity and not the RelatedPlaceRefOrValue class itself.
riskAssessmentResult	A RiskAssessmentResult. This resource reflects the risk assessment.
status	A String. The status of the risk assessment, such as Succeeded, In Progress, Failed, output from the task.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'ProductOrderRiskAssessment' resource object.

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```
{
    "@type": "ProductOrderRiskAssessment",
    "status": "Completed",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "CouponType",
            "valueType": "string",
            "value": "Discount21"
        }
    ],
    "productOrder": {
        "id": "267734",
        "href": "https://mycsp.com:8080/tmf-
api/productOrder/v5/productOrder/267734",
        "@referredType": "ProductOrder",
        "@type": "ProductOrderRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 16,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "IDConfidenceRisk"
            },
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "PaymentMethodRisk"
            },
            {
                "@type": "RiskScore",
                "score": 15,
                "riskName": "FraudRisk"
            },
            {
                "@type": "RiskScore",
                "score": 14,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 7,
                "riskName": "CreditGamingRisk"
            }
```


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```
        ]
    }
}
```

Notification Resource Models

15 notifications are defined for this API.

Notifications related to ProductOfferingRiskAssessment:

- Create Event
- Delete Event
- Status Change Event

Notifications related to PartyRoleRiskAssessment:

- Create Event
- Delete Event
- Status Change Event

Notifications related to PartyRoleProductOfferingRiskAssessment:

- Create Event
- Delete Event
- Status Change Event

Notifications related to ShoppingCartRiskAssessment:

- Create Event
- Delete Event
- Status Change Event

Notifications related to ProductOrderRiskAssessment:

- Create Event
- Delete Event
- Status Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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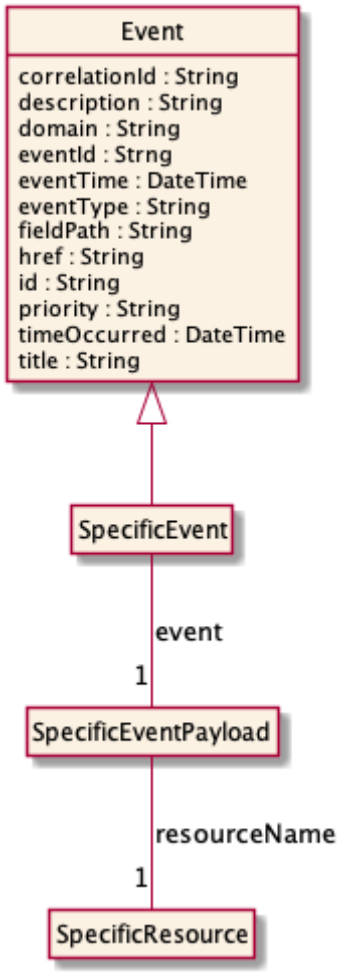


Figure 21 Notification Pattern

ProductOfferingRiskAssessment

Create Event

Message example for ProductOfferingRiskAssessmentCreateEvent event

```
Content-Type: application/json

{
    "correlationId": "24fab9e4-b138",
    "description": "ProductOfferingRiskAssessmentCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4c11-ba1e-484b004ba625",
    "eventTime": "2023-03-05T12:42:24.314Z",
    "eventType": "ProductOfferingRiskAssessmentCreateEvent",
    "priority": "5",
    "timeOcurred": "2023-03-05T12:42:20.095Z",
    "title": "ProductOfferingRiskAssessmentCreateEvent",
    "event": {
        "productOfferingRiskAssessment": {
            "@type": "ProductOfferingRiskAssessment",
            "id": "777666345666",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/productOfferingRiskAssessment/777666345666",
```

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```
            "status": "InProgress",
            "characteristic": [
                {
                    "@type": "StringCharacteristic",
                    "name": "Bandwidth",
                    "valueType": "string",
                    "value": "500MBps"
                }
            ],
            "productOffering": {
                "id": "7656",
                "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
                "name": "Home Gateway",
                "@referredType": "ProductOffering",
                "@type": "ProductOfferingRef"
            },
            "relatedPlace": {
                "@type": "RelatedPlaceRefOrValue",
                "role": "home address",
                "place": {
                    "id": "234546467567334",
                    "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                    "@type": "PlaceRef",
                    "@referredType": "GeographicAddress"
                }
            }
        }
    },
    "reportingSystem": {
        "id": "480",
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProductOfferingRiskAssessmentCreateEvent"
}
```


Delete Event

Message example for ProductOfferingRiskAssessmentDeleteEvent event


```
Content-Type: application/json

{
    "correlationId": "47716312-3655",
    "description": "ProductOfferingRiskAssessmentDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "48b7-a3ca-0359f04a09ea",
    "eventTime": "2023-03-05T12:42:24.315Z",
    "eventType": "ProductOfferingRiskAssessmentDeleteEvent",
    "priority": "3",
```


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```
    "timeOcurred": "2023-03-05T12:42:18.773Z",
    "title": "ProductOfferingRiskAssessmentDeleteEvent",
    "event": {
        "productOfferingRiskAssessment": {
            "@type": "ProductOfferingRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/ProductOfferingRiskAssessment/777666121888"
        }
    },
    "reportingSystem": {
        "id": "480",
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProductOfferingRiskAssessmentDeleteEvent"
}
```


Status Change Event

Message example for ProductOfferingRiskAssessmentStatusChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "9dc565e2-b9ef",
    "description": "ProductOfferingRiskAssessmentStatusChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4c64-b45b-f34e242c5a69",
    "eventTime": "2022-08-25T12:27:33.472Z",
    "eventType": "ProductOfferingRiskAssessmentStatusChangeEvent",
    "priority": "2",
    "timeOcurred": "2022-08-25T12:27:29.006Z",
    "title": "ProductOfferingRiskAssessmentStatusChangeEvent",
    "event": {
        "productOfferingRiskAssessment": {
            "@type": "ProductOfferingRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/ProductOfferingRiskAssessment/777666121888",
            "status": "Completed"
        }
    },
    "reportingSystem": {
        "id": "92",
        "name": "APP-983",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "706",
        "name": "APP-626",
```


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```
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProductOfferingRiskAssessmentStatusChangeEvent"
}
```

PartyRoleRiskAssessment

Create Event

Message example for PartyRoleRiskAssessmentCreateEvent event

```
Content-Type: application/json

{
    "correlationId": "24fab9e4-b138",
    "description": "PartyRoleRiskAssessmentCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4c11-ba1e-484b004ba625",
    "eventTime": "2023-03-05T12:42:24.314Z",
    "eventType": "PartyRoleRiskAssessmentCreateEvent",
    "priority": "5",
    "timeOcurred": "2023-03-05T12:42:20.095Z",
    "title": "PartyRoleRiskAssessmentCreateEvent",
    "event": {
        "partyRoleRiskAssessment": {
            "@type": "PartyRoleRiskAssessment",
            "id": "777666444555",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/partyRoleRiskAssessment/777666444555",
            "status": "InProgress",
            "partyRole": {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "role": "potential buyer",
                "partyOrPartyRole": {
                    "id": "9866",
                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
                    "name": "Sandy Smith",
                    "@type": "PartyRef",
                    "@referredType": "Individual"
                }
            },
            "relatedPlace": {
                "@type": "RelatedPlaceRefOrValue",
                "role": "home address",
                "place": {
                    "id": "234546467567334",
                    "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                    "@type": "PlaceRef",
                    "@referredType": "GeographicAddress"
                }
            }
        }
    },
    "reportingSystem": {
        "id": "480",
```

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```
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PartyRoleRiskAssessmentCreateEvent"
}
```


Delete Event

Message example for PartyRoleRiskAssessmentDeleteEvent event


```
Content-Type: application/json

{
    "correlationId": "47716312-3655",
    "description": "PartyRoleRiskAssessmentDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "48b7-a3ca-0359f04a09ea",
    "eventTime": "2023-03-05T12:42:24.315Z",
    "eventType": "PartyRoleRiskAssessmentDeleteEvent",
    "priority": "3",
    "timeOcurred": "2023-03-05T12:42:18.773Z",
    "title": "PartyRoleRiskAssessmentDeleteEvent",
    "event": {
        "partyRoleRiskAssessment": {
            "@type": "PartyRoleRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/PartyRoleRiskAssessment/777666121888"
        }
    },
    "reportingSystem": {
        "id": "480",
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PartyRoleRiskAssessmentDeleteEvent"
}
```


Status Change Event

Message example for PartyRoleRiskAssessmentStatusChangeEvent event

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```
Content-Type: application/json

{
    "correlationId": "9dc565e2-b9ef",
    "description": "PartyRoleRiskAssessmentStatusChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4c64-b45b-f34e242c5a69",
    "eventTime": "2022-08-25T12:27:33.472Z",
    "eventType": "PartyRoleRiskAssessmentStatusChangeEvent",
    "priority": "2",
    "timeOcurred": "2022-08-25T12:27:29.006Z",
    "title": "PartyRoleRiskAssessmentStatusChangeEvent",
    "event": {
        "partyRoleRiskAssessment": {
            "@type": "PartyRoleRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/PartyRoleRiskAssessment/777666121888",
            "status": "Completed"
        }
    },
    "reportingSystem": {
        "id": "92",
        "name": "APP-983",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "706",
        "name": "APP-626",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PartyRoleRiskAssessmentStatusChangeEvent"
}
```

PartyRoleProductOfferingRiskAssessment

Create Event

Message example for PartyRoleProductOfferingRiskAssessmentCreateEvent event

```
Content-Type: application/json

{
    "correlationId": "24fab9e4-b138",
    "description": "PartyRoleProductOfferingRiskAssessmentCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4c11-ba1e-484b004ba625",
    "eventTime": "2023-03-05T12:42:24.314Z",
    "eventType": "PartyRoleProductOfferingRiskAssessmentCreateEvent",
    "priority": "5",
    "timeOcurred": "2023-03-05T12:42:20.095Z",
    "title": "PartyRoleProductOfferingRiskAssessmentCreateEvent",
    "event": {
        "partyRoleProductOfferingRiskAssessment": {
            "@type": "PartyRoleProductOfferingRiskAssessment",
```

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```
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/partyRoleProductOfferingRiskAssessment/777666121888",
            "status": "InProgress",
            "characteristic": [
                {
                    "@type": "StringCharacteristic",
                    "name": "Bandwidth",
                    "valueType": "string",
                    "value": "500MBps"
                }
            ],
            "partyRole": {
                "@type": "RelatedPartyRefOrPartyRoleRef",
                "role": "potential buyer",
                "partyOrPartyRole": {
                    "id": "9866",
                    "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
                    "name": "Sandy Smith",
                    "@type": "PartyRef",
                    "@referredType": "Individual"
                }
            },
            "productOffering": {
                "id": "7656",
                "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
                "name": "Home Gateway",
                "@referredType": "ProductOffering",
                "@type": "ProductOfferingRef"
            },
            "relatedPlace": {
                "@type": "RelatedPlaceRefOrValue",
                "role": "home address",
                "place": {
                    "id": "234546467567334",
                    "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                    "@type": "PlaceRef",
                    "@referredType": "GeographicAddress"
                }
            }
        }
    },
    "reportingSystem": {
        "id": "480",
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PartyRoleProductOfferingRiskAssessmentCreateEvent"
}
```


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Delete Event

Message example for PartyRoleProductOfferingRiskAssessmentDeleteEvent event

```
Content-Type: application/json

{
    "correlationId": "47716312-3655",
    "description": "PartyRoleProductOfferingRiskAssessmentDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "48b7-a3ca-0359f04a09ea",
    "eventTime": "2023-03-05T12:42:24.315Z",
    "eventType": "PartyRoleProductOfferingRiskAssessmentDeleteEvent",
    "priority": "3",
    "timeOcurred": "2023-03-05T12:42:18.773Z",
    "title": "PartyRoleProductOfferingRiskAssessmentDeleteEvent",
    "event": {
        "partyRoleProductOfferingRiskAssessment": {
            "@type": "PartyRoleProductOfferingRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/PartyRoleProductOfferingRiskAssessment/777666121888"
        }
    },
    "reportingSystem": {
        "id": "480",
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PartyRoleProductOfferingRiskAssessmentDeleteEvent"
}
```

Status Change Event

Message example for PartyRoleProductOfferingRiskAssessmentStatusChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "9dc565e2-b9ef",
    "description": "PartyRoleProductOfferingRiskAssessmentStatusChangeEvent
illustration",
    "domain": "Commercial",
    "eventId": "4c64-b45b-f34e242c5a69",
    "eventTime": "2022-08-25T12:27:33.472Z",
    "eventType": "PartyRoleProductOfferingRiskAssessmentStatusChangeEvent",
    "priority": "2",
    "timeOcurred": "2022-08-25T12:27:29.006Z",
    "title": "PartyRoleProductOfferingRiskAssessmentStatusChangeEvent",
    "event": {
```

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```
        "partyRoleProductOfferingRiskAssessment": {
            "@type": "PartyRoleProductOfferingRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/PartyRoleProductOfferingRiskAssessment/777666121888",
            "status": "Completed"
        }
    },
    "reportingSystem": {
        "id": "92",
        "name": "APP-983",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "706",
        "name": "APP-626",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PartyRoleProductOfferingRiskAssessmentStatusChangeEvent"
}
```


ShoppingCartRiskAssessment

Create Event

Message example for ShoppingCartRiskAssessmentCreateEvent event


```
Content-Type: application/json

{
    "correlationId": "24fab9e4-b138",
    "description": "ShoppingCartRiskAssessmentCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4c11-ba1e-484b004ba625",
    "eventTime": "2023-03-05T12:42:24.314Z",
    "eventType": "ShoppingCartRiskAssessmentCreateEvent",
    "priority": "5",
    "timeOcurred": "2023-03-05T12:42:20.095Z",
    "title": "ShoppingCartRiskAssessmentCreateEvent",
    "event": {
        "shoppingCartRiskAssessment": {
            "@type": "ShoppingCartRiskAssessment",
            "id": "777666345777",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/shoppingCartRiskAssessment/777666345777",
            "status": "InProgress",
            "characteristic": [
                {
                    "@type": "StringCharacteristic",
                    "name": "ContactPersonName",
                    "valueType": "string",
                    "value": "David Alexander"
                }
            ],
            "shoppingCart": {
                "id": "1203",
```


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```
                "href": "https://mycsp.com:8080/tmf-
api/shoppingCart/v5/shoppingCart/1203",
                "@referredType": "ShoppingCart",
                "@type": "ShoppingCartRef"
            },
            "relatedPlace": {
                "@type": "RelatedPlaceRefOrValue",
                "role": "home address",
                "place": {
                    "id": "234546467567334",
                    "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                    "@type": "PlaceRef",
                    "@referredType": "GeographicAddress"
                }
            }
        }
    },
    "reportingSystem": {
        "id": "480",
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ShoppingCartRiskAssessmentCreateEvent"
}
```


Delete Event

Message example for ShoppingCartRiskAssessmentDeleteEvent event


```
Content-Type: application/json

{
    "correlationId": "47716312-3655",
    "description": "ShoppingCartRiskAssessmentDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "48b7-a3ca-0359f04a09ea",
    "eventTime": "2023-03-05T12:42:24.315Z",
    "eventType": "ShoppingCartRiskAssessmentDeleteEvent",
    "priority": "3",
    "timeOcurred": "2023-03-05T12:42:18.773Z",
    "title": "ShoppingCartRiskAssessmentDeleteEvent",
    "event": {
        "shoppingCartRiskAssessment": {
            "@type": "ShoppingCartRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/ShoppingCartRiskAssessment/777666121888"
        }
    },
    "reportingSystem": {
        "id": "480",
```


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```
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ShoppingCartRiskAssessmentDeleteEvent"
}
```


Status Change Event

Message example for ShoppingCartRiskAssessmentStatusChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "9dc565e2-b9ef",
    "description": "ShoppingCartRiskAssessmentStatusChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "4c64-b45b-f34e242c5a69",
    "eventTime": "2022-08-25T12:27:33.472Z",
    "eventType": "ShoppingCartRiskAssessmentStatusChangeEvent",
    "priority": "2",
    "timeOcurred": "2022-08-25T12:27:29.006Z",
    "title": "ShoppingCartRiskAssessmentStatusChangeEvent",
    "event": {
        "shoppingCartRiskAssessment": {
            "@type": "ShoppingCartRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/shoppingCartRiskAssessment/777666121888",
            "status": "Completed"
        }
    },
    "reportingSystem": {
        "id": "92",
        "name": "APP-983",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "706",
        "name": "APP-626",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ShoppingCartRiskAssessmentStatusChangeEvent"
}
```


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ProductOrderRiskAssessment

Create Event

Message example for ProductOrderRiskAssessmentCreateEvent event

```
Content-Type: application/json

{
    "correlationId": "24fab9e4-b138",
    "description": "ProductOrderRiskAssessmentCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "4c11-ba1e-484b004ba625",
    "eventTime": "2023-03-05T12:42:24.314Z",
    "eventType": "ProductOrderRiskAssessmentCreateEvent",
    "priority": "5",
    "timeOcurred": "2023-03-05T12:42:20.095Z",
    "title": "ProductOrderRiskAssessmentCreateEvent",
    "event": {
        "productOrderRiskAssessment": {
            "@type": "ProductOrderRiskAssessment",
            "id": "777666121999",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/productOrderRiskAssessment/777666121999",
            "status": "InProgress",
            "characteristic": [
                {
                    "@type": "StringCharacteristic",
                    "name": "CouponType",
                    "valueType": "string",
                    "value": "Discount21"
                }
            ],
            "productOrder": {
                "id": "267734",
                "href": "https://mycsp.com:8080/tmf-
api/productOrder/v5/productOrder/267734",
                "@referredType": "ProductOrder",
                "@type": "ProductOrderRef"
            },
            "relatedPlace": {
                "@type": "RelatedPlaceRefOrValue",
                "role": "home address",
                "place": {
                    "id": "234546467567334",
                    "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                    "@type": "PlaceRef",
                    "@referredType": "GeographicAddress"
                }
            }
        }
    },
    "reportingSystem": {
        "id": "480",
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
```

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```
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProductOrderRiskAssessmentCreateEvent"
}
```

Delete Event

Message example for ProductOrderRiskAssessmentDeleteEvent event

```
Content-Type: application/json

{
    "correlationId": "47716312-3655",
    "description": "ProductOrderRiskAssessmentDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "48b7-a3ca-0359f04a09ea",
    "eventTime": "2023-03-05T12:42:24.315Z",
    "eventType": "ProductOrderRiskAssessmentDeleteEvent",
    "priority": "3",
    "timeOcurred": "2023-03-05T12:42:18.773Z",
    "title": "ProductOrderRiskAssessmentDeleteEvent",
    "event": {
        "productOrderRiskAssessment": {
            "@type": "ProductOrderRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/ProductOrderRiskAssessment/777666121888"
        }
    },
    "reportingSystem": {
        "id": "480",
        "name": "APP-21",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "416",
        "name": "APP-400",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProductOrderRiskAssessmentDeleteEvent"
}
```

Status Change Event

Message example for ProductOrderRiskAssessmentStatusChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "9dc565e2-b9ef",
    "description": "ProductOrderRiskAssessmentStatusChangeEvent illustration",
```

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```
    "domain": "Commercial",
    "eventId": "4c64-b45b-f34e242c5a69",
    "eventTime": "2022-08-25T12:27:33.472Z",
    "eventType": "ProductOrderRiskAssessmentStatusChangeEvent",
    "priority": "2",
    "timeOcurred": "2022-08-25T12:27:29.006Z",
    "title": "ProductOrderRiskAssessmentStatusChangeEvent",
    "event": {
        "productOrderRiskAssessment": {
            "@type": "ProductOrderRiskAssessment",
            "id": "777666121888",
            "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/ProductOrderRiskAssessment/777666121888",
            "status": "Completed"
        }
    },
    "reportingSystem": {
        "id": "92",
        "name": "APP-983",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "706",
        "name": "APP-626",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "ProductOrderRiskAssessmentStatusChangeEvent"
}
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on ProductOfferingRiskAssessment

List or find ProductOfferingRiskAssessment objects

`GET /productOfferingRiskAssessment?fields=…`

Description

This operation list ProductOfferingRiskAssessment entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving multiple tasks for party role risk assessment.

Request

```
GET /productOfferingRiskAssessment
Content-Type: application/json
```

Response

```
200
```

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```
[
    {
        "@type": "ProductOfferingRiskAssessment",
        "id": "777666345666",
        "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/productOfferingRiskAssessment/777666345666",
        "status": "InProgress",
        "characteristic": [
            {
                "@type": "StringCharacteristic",
                "name": "Bandwidth",
                "valueType": "string",
                "value": "500MBps"
            }
        ],
        "productOffering": {
            "id": "7656",
            "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
            "name": "Home Gateway",
            "@referredType": "ProductOffering",
            "@type": "ProductOfferingRef"
        },
        "relatedPlace": {
            "@type": "RelatedPlaceRefOrValue",
            "role": "home address",
            "place": {
                "id": "234546467567334",
                "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                "@type": "PlaceRef",
                "@referredType": "GeographicAddress"
            }
        }
    }
]
```


Retrieves a ProductOfferingRiskAssessment by ID

`GET /productOfferingRiskAssessment/{id}?fields=…`

Description

This operation retrieves a ProductOfferingRiskAssessment entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving a specific task for product offering risk assessment.

Request

```
GET /productOfferingRiskAssessment/777666345666
Content-Type: application/json
```

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Response

```
200

{
    "@type": "ProductOfferingRiskAssessment",
    "id": "777666345666",
    "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/productOfferingRiskAssessment/777666345666",
    "status": "InProgress",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "Bandwidth",
            "valueType": "string",
            "value": "500MBps"
        }
    ],
    "productOffering": {
        "id": "7656",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
        "name": "Home Gateway",
        "@referredType": "ProductOffering",
        "@type": "ProductOfferingRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
}
```

Creates a ProductOfferingRiskAssessment

`POST /productOfferingRiskAssessment?fields=…`

Description

This operation creates a ProductOfferingRiskAssessment entity.

Mandatory Attributes

Mandatory Attributes	Rule
productOffering	
@type	

Usage samples

Here is an example of a request for creating a specific task for product offering risk

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assessment.

Request

```
POST /productOfferingRiskAssessment
Content-Type: application/json

{
    "@type": "ProductOfferingRiskAssessment",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "Bandwidth",
            "valueType": "string",
            "value": "500MBps"
        }
    ],
    "productOffering": {
        "id": "7656",
        "@referredType": "ProductOffering",
        "@type": "ProductOfferingRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
}
```

Response

```
201

{
    "@type": "ProductOfferingRiskAssessment",
    "status": "Completed",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "Bandwidth",
            "valueType": "string",
            "value": "500MBps"
        }
    ],
    "productOffering": {
        "id": "7656",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
        "name": "Home Gateway",
        "@referredType": "ProductOffering",
        "@type": "ProductOfferingRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
```

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```
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 19,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "@type": "RiskScore",
                "score": 20,
                "riskName": "FraudRisk"
            },
            {
                "@type": "RiskScore",
                "score": 7,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 5,
                "riskName": "CreditGamingRisk"
            }
        ]
    }
}
```


Deletes a ProductOfferingRiskAssessment

`DELETE /productOfferingRiskAssessment/{id}`

Description

This operation deletes a ProductOfferingRiskAssessment entity.

Operations on PartyRoleRiskAssessment

Retrieves a PartyRoleRiskAssessment by ID

`GET /partyRoleRiskAssessment/{id}?fields=…`

Description

This operation retrieves a PartyRoleRiskAssessment entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

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Here is an example of a request for retrieving a specific task for party role risk assessment.

Request

```
GET /partyRoleRiskAssessment/777666444555
Content-Type: application/json
```

Response

```
200

{
    "@type": "PartyRoleRiskAssessment",
    "id": "777666444555",
    "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/partyRoleRiskAssessment/777666444555",
    "status": "InProgress",
    "partyRole": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "potential buyer",
        "partyOrPartyRole": {
            "id": "9866",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
        }
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
}
```

List or find PartyRoleRiskAssessment objects

`GET /partyRoleRiskAssessment?fields=…`

Description

This operation list PartyRoleRiskAssessment entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving multiple tasks for party role risk assessment.

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Request

```
GET /partyRoleRiskAssessment?status=InProgress
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "PartyRoleRiskAssessment",
        "id": "777666444555",
        "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/partyRoleRiskAssessment/777666444555",
        "status": "InProgress",
        "partyRole": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "potential buyer",
            "partyOrPartyRole": {
                "id": "9866",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
                "name": "Sandy Smith",
                "@type": "PartyRef",
                "@referredType": "Individual"
            }
        },
        "relatedPlace": {
            "@type": "RelatedPlaceRefOrValue",
            "role": "home address",
            "place": {
                "id": "234546467567334",
                "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                "@type": "PlaceRef",
                "@referredType": "GeographicAddress"
            }
        }
    }
]
```

Creates a PartyRoleRiskAssessment

POST /partyRoleRiskAssessment?fields=…

Description

This operation creates a PartyRoleRiskAssessment entity.

Mandatory Attributes

Mandatory Attributes	Rule
partyRole	

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Mandatory Attributes	Rule
partyRole.@type	
partyRole.role	
@type	

Usage samples

Here is an example of a request for creating a specific task for party role risk assessment.

Request

```
POST /partyRoleRiskAssessment
Content-Type: application/json

{
    "@type": "PartyRoleRiskAssessment",
    "partyRole": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "potential buyer",
        "partyOrPartyRole": {
            "id": "9866",
            "@type": "PartyRef",
            "@referredType": "Individual"
        }
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
}
```

Response

```
201

{
    "@type": "PartyRoleRiskAssessment",
    "status": "Completed",
    "partyRole": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "potential buyer",
        "partyOrPartyRole": {
            "id": "9866",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
        }
    },
```

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```
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 65,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "IDConfidenceRisk"
            },
            {
                "@type": "RiskScore",
                "score": 80,
                "riskName": "FraudRisk"
            },
            {
                "@type": "RiskScore",
                "score": 20,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 14,
                "riskName": "CreditGamingRisk"
            }
        ]
    }
}
```


Deletes a PartyRoleRiskAssessment

`DELETE /partyRoleRiskAssessment/{id}`

Description

This operation deletes a PartyRoleRiskAssessment entity.

Operations on PartyRoleProductOfferingRiskAssessment

List or find PartyRoleProductOfferingRiskAssessment objects

`GET /partyRoleProductOfferingRiskAssessment?fields=…`

Description

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This operation list PartyRoleProductOfferingRiskAssessment entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving multiple tasks for party role product offering risk assessment.

Request

```
GET /partyRoleProductOfferingRiskAssessment
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "PartyRoleProductOfferingRiskAssessment",
        "id": "777666121888",
        "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/partyRoleProductOfferingRiskAssessment/777666121888",
        "status": "InProgress",
        "characteristic": [
            {
                "@type": "StringCharacteristic",
                "name": "Bandwidth",
                "valueType": "string",
                "value": "500MBps"
            }
        ],
        "partyRole": {
            "@type": "RelatedPartyRefOrPartyRoleRef",
            "role": "potential buyer",
            "partyOrPartyRole": {
                "id": "9866",
                "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
                "name": "Sandy Smith",
                "@type": "PartyRef",
                "@referredType": "Individual"
            }
        },
        "productOffering": {
            "id": "7656",
            "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
            "name": "Home Gateway",
            "@referredType": "ProductOffering",
            "@type": "ProductOfferingRef"
        },
        "relatedPlace": {
            "@type": "RelatedPlaceRefOrValue",
            "role": "home address",
            "place": {
```

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```
                "id": "234546467567334",
                "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                "@type": "PlaceRef",
                "@referredType": "GeographicAddress"
            }
        }
    }
]
```

Retrieves a PartyRoleProductOfferingRiskAssessment by ID

`GET /partyRoleProductOfferingRiskAssessment/{id}?fields=…`

Description

This operation retrieves a PartyRoleProductOfferingRiskAssessment entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving a specific task for party role product offering risk assessment.

Request

```
GET /partyRoleProductOfferingRiskAssessment/777666121888
Content-Type: application/json
```

Response

```
200

{
    "@type": "PartyRoleProductOfferingRiskAssessment",
    "id": "777666121888",
    "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/partyRoleProductOfferingRiskAssessment/777666121888",
    "status": "InProgress",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "Bandwidth",
            "valueType": "string",
            "value": "500MBps"
        }
    ],
    "partyRole": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "potential buyer",
        "partyOrPartyRole": {
            "id": "9866",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
```

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```
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
        }
    },
    "productOffering": {
        "id": "7656",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
        "name": "Home Gateway",
        "@referredType": "ProductOffering",
        "@type": "ProductOfferingRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
}
```


Creates a PartyRoleProductOfferingRiskAssessment

`POST /partyRoleProductOfferingRiskAssessment?fields=…`

Description

This operation creates a PartyRoleProductOfferingRiskAssessment entity.

Mandatory Attributes

Mandatory Attributes	Rule
partyRole	
partyRole.@type	
partyRole.role	
productOffering	
@type	

Usage samples

Here is an example of a request for creating a specific task for party role product offering risk assessment.

Request

```
POST /partyRoleProductOfferingRiskAssessment
Content-Type: application/json
```

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```
{
    "@type": "PartyRoleProductOfferingRiskAssessment",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "Bandwidth",
            "valueType": "string",
            "value": "500MBps"
        }
    ],
    "partyRole": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "potential buyer",
        "partyOrPartyRole": {
            "id": "9866",
            "@type": "PartyRef",
            "@referredType": "Individual"
        }
    },
    "productOffering": {
        "id": "7656",
        "@referredType": "ProductOffering",
        "@type": "ProductOfferingRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
}
```


Response


```
201

{
    "@type": "PartyRoleProductOfferingRiskAssessment",
    "status": "Completed",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "Bandwidth",
            "valueType": "string",
            "value": "500MBps"
        }
    ],
    "partyRole": {
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "role": "potential buyer",
        "partyOrPartyRole": {
            "id": "9866",
            "href": "https://mycsp.com:8080/tmf-
api/partyManagement/v5/individual/9866",
            "name": "Sandy Smith",
            "@type": "PartyRef",
            "@referredType": "Individual"
```


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```
        }
    },
    "productOffering": {
        "id": "7656",
        "href": "https://mycsp.com:8080/tmf-
api/productCatalogManagement/v5/productOffering/7656",
        "name": "Home Gateway",
        "@referredType": "ProductOffering",
        "@type": "ProductOfferingRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 21,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "IDConfidenceRisk"
            },
            {
                "@type": "RiskScore",
                "score": 32,
                "riskName": "FraudRisk"
            },
            {
                "@type": "RiskScore",
                "score": 8,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 16,
                "riskName": "CreditGamingRisk"
            }
        ]
    }
}
```


Deletes a PartyRoleProductOfferingRiskAssessment

`DELETE /partyRoleProductOfferingRiskAssessment/{id}`

Description

This operation deletes a PartyRoleProductOfferingRiskAssessment entity.

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Operations on ShoppingCartRiskAssessment

Retrieves a ShoppingCartRiskAssessment by ID

`GET /shoppingCartRiskAssessment/{id}?fields=…`

Description

This operation retrieves a ShoppingCartRiskAssessment entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving a specific task for shopping cart risk assessment.

Request

```
GET /shoppingCartRiskAssessment/777666123777
Content-Type: application/json
```

Response

```
200

{
    "@type": "ShoppingCartRiskAssessment",
    "id": "777666345777",
    "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/shoppingCartRiskAssessment/777666345777",
    "status": "InProgress",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "ContactPersonName",
            "valueType": "string",
            "value": "David Alexander"
        }
    ],
    "shoppingCart": {
        "id": "1203",
        "href": "https://mycsp.com:8080/tmf-api/shoppingCart/v5/shoppingCart/1203",
        "@referredType": "ShoppingCart",
        "@type": "ShoppingCartRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
```

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```
    }
}
```

List or find ShoppingCartRiskAssessment objects

`GET /shoppingCartRiskAssessment?fields=…`

Description

This operation list ShoppingCartRiskAssessment entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving multiple tasks for party role risk assessment.

Request

```
GET /shoppingCartRiskAssessment?status=InProgress
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "ShoppingCartRiskAssessment",
        "id": "777666345777",
        "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/shoppingCartRiskAssessment/777666345777",
        "status": "InProgress",
        "characteristic": [
            {
                "@type": "StringCharacteristic",
                "name": "ContactPersonName",
                "valueType": "string",
                "value": "David Alexander"
            }
        ],
        "shoppingCart": {
            "id": "1203",
            "href": "https://mycsp.com:8080/tmf-
api/shoppingCart/v5/shoppingCart/1203",
            "@referredType": "ShoppingCart",
            "@type": "ShoppingCartRef"
        },
        "relatedPlace": {
            "@type": "RelatedPlaceRefOrValue",
            "role": "home address",
            "place": {
                "id": "234546467567334",
                "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
```

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```
                "@type": "PlaceRef",
                "@referredType": "GeographicAddress"
            }
        }
    }
]
```

Creates a ShoppingCartRiskAssessment

`POST /shoppingCartRiskAssessment?fields=…`

Description

This operation creates a ShoppingCartRiskAssessment entity.

Mandatory Attributes

Mandatory Attributes	Rule
shoppingCart	
@type	

Usage samples

Here is an example of a request for creating a specific task for shopping cart risk assessment.

Request

```
POST /shoppingCartRiskAssessment
Content-Type: application/json

{
    "@type": "ShoppingCartRiskAssessment",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "ContactPersonName",
            "valueType": "string",
            "value": "David Alexander"
        }
    ],
    "shoppingCart": {
        "id": "1203",
        "@referredType": "ShoppingCart",
        "@type": "ShoppingCartRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
```

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```
}
```

Response

```
201

{
    "@type": "ShoppingCartRiskAssessment",
    "status": "Completed",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "ContactPersonName",
            "valueType": "string",
            "value": "David Alexander"
        }
    ],
    "shoppingCart": {
        "id": "1203",
        "href": "https://mycsp.com:8080/tmf-api/shoppingCart/v5/shoppingCart/1203",
        "@referredType": "ShoppingCart",
        "@type": "ShoppingCartRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 12,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "IDConfidenceRisk"
            },
            {
                "@type": "RiskScore",
                "score": 10,
                "riskName": "FraudRisk"
            },
            {
                "@type": "RiskScore",
                "score": 13,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 9,
```

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```
                "riskName": "CreditGamingRisk"
            }
        ]
    }
}
```

Deletes a ShoppingCartRiskAssessment

`DELETE /shoppingCartRiskAssessment/{id}`

Description

This operation deletes a ShoppingCartRiskAssessment entity.

Operations on ProductOrderRiskAssessment

List or find ProductOrderRiskAssessment objects

`GET /productOrderRiskAssessment?fields=…`

Description

This operation list ProductOrderRiskAssessment entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving multiple tasks for party role risk assessment.

Request

```
GET /productOrderRiskAssessment
Content-Type: application/json
```

Response

```
200

[
    {
        "@type": "ProductOrderRiskAssessment",
        "id": "777666121999",
        "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/productOrderRiskAssessment/777666121999",
        "status": "InProgress",
        "characteristic": [
            {
                "@type": "StringCharacteristic",
                "name": "CouponType",
                "valueType": "string",
                "value": "Discount21"
            }
```

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```
        ],
        "productOrder": {
            "id": "267734",
            "href": "https://mycsp.com:8080/tmf-
api/productOrder/v5/productOrder/267734",
            "@referredType": "ProductOrder",
            "@type": "ProductOrderRef"
        },
        "relatedPlace": {
            "@type": "RelatedPlaceRefOrValue",
            "role": "home address",
            "place": {
                "id": "234546467567334",
                "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
                "@type": "PlaceRef",
                "@referredType": "GeographicAddress"
            }
        }
    }
]
```


Retrieves a ProductOrderRiskAssessment by ID

`GET /productOrderRiskAssessment/{id}?fields=…`

Description

This operation retrieves a ProductOrderRiskAssessment entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here is an example of a request for retrieving a specific task for party role risk assessment.

Request

```
GET /productOrderRiskAssessment/777666121999
Content-Type: application/json
```

Response


```
200

{
    "@type": "ProductOrderRiskAssessment",
    "id": "777666121999",
    "href": "https://mycsp.com:8080/tmf-
api/riskManagement/v5/productOrderRiskAssessment/777666121999",
    "status": "InProgress",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "CouponType",
```


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```
            "valueType": "string",
            "value": "Discount21"
        }
    ],
    "productOrder": {
        "id": "267734",
        "href": "https://mycsp.com:8080/tmf-
api/productOrder/v5/productOrder/267734",
        "@referredType": "ProductOrder",
        "@type": "ProductOrderRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
}
```

Creates a ProductOrderRiskAssessment

`POST /productOrderRiskAssessment?fields=…`

Description

This operation creates a ProductOrderRiskAssessment entity.

Mandatory Attributes

Mandatory Attributes	Rule
productOrder	
@type	

Usage samples

Here is an example of a request for creating a specific task for product order risk assessment.

Request

```
POST /productOrderRiskAssessment
Content-Type: application/json

{
    "@type": "ProductOrderRiskAssessment",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "CouponType",
            "valueType": "string",
```

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```
            "value": "Discount21"
        }
    ],
    "productOrder": {
        "id": "267734",
        "@referredType": "ProductOrder",
        "@type": "ProductOrderRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    }
}
```


Response


```
201

{
    "@type": "ProductOrderRiskAssessment",
    "status": "Completed",
    "characteristic": [
        {
            "@type": "StringCharacteristic",
            "name": "CouponType",
            "valueType": "string",
            "value": "Discount21"
        }
    ],
    "productOrder": {
        "id": "267734",
        "href": "https://mycsp.com:8080/tmf-
api/productOrder/v5/productOrder/267734",
        "@referredType": "ProductOrder",
        "@type": "ProductOrderRef"
    },
    "relatedPlace": {
        "@type": "RelatedPlaceRefOrValue",
        "role": "home address",
        "place": {
            "id": "234546467567334",
            "href": "https://mycsp.com:8080/tmf-
api/geographicAddress/v5/geographicAddress/234546467567334",
            "@type": "PlaceRef",
            "@referredType": "GeographicAddress"
        }
    },
    "riskAssessmentResult": {
        "@type": "RiskAssessmentResult",
        "overallScore": 16,
        "validFor": {
            "startDateTime": "2017-08-23T00:00:22.920Z",
            "endDateTime": "2018-03-25T00:00:22.920Z"
        },
        "score": [
```


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```
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "IDConfidenceRisk"
            },
            {
                "@type": "RiskScore",
                "score": 0,
                "riskName": "PaymentMethodRisk"
            },
            {
                "@type": "RiskScore",
                "score": 15,
                "riskName": "FraudRisk"
            },
            {
                "@type": "RiskScore",
                "score": 14,
                "riskName": "BadPaymentRisk"
            },
            {
                "@type": "RiskScore",
                "score": 7,
                "riskName": "CreditGamingRisk"
            }
        ]
    }
}
```


Deletes a ProductOrderRiskAssessment

`DELETE /productOrderRiskAssessment/{id}`

Description

This operation deletes a ProductOrderRiskAssessment entity.

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

Release History

Release Number	Date	Release led by:	Description
Release 0.1	24/09/2019	Moshe Zolotov Amdocs Moshe.Zolotov@amdocs.com	Initial work-in-progress iteration of draft.
Release 1.0	25/03/2020	Jonathan Goldberg Amdocs Jonathan.Goldberg@Amdocs.com	Alignment with design guidelines, correcting examples
Version 5.0	09/03/2024	Jonathan Goldberg Amdocs Jonathan.Goldberg@Amdocs.com	Alignment with design guidelines, correcting examples

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